Hyperconvex representations and Richardson varieties

Advised by Emily Dumas

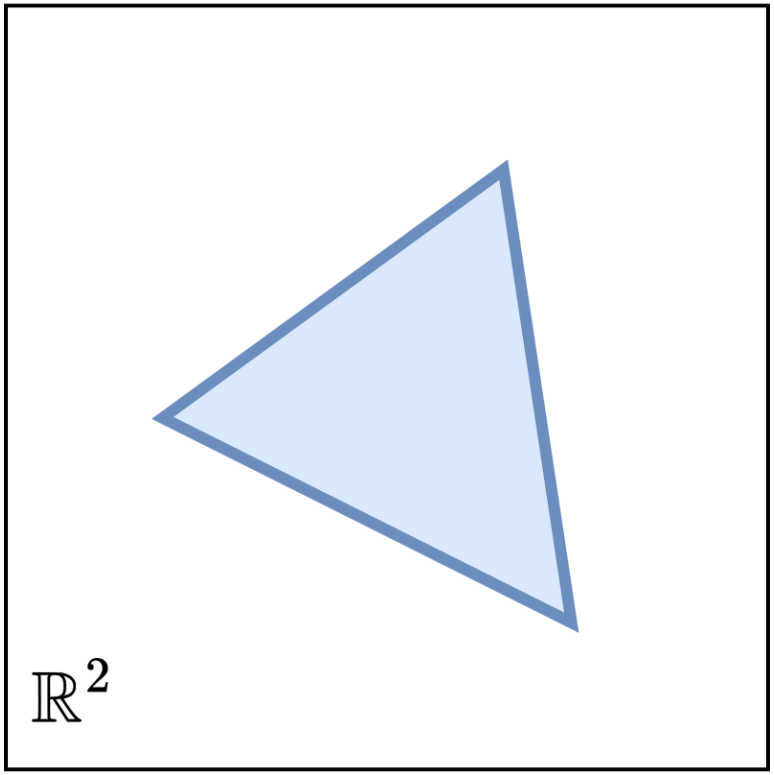
Jennifer Lynn Vaccaro

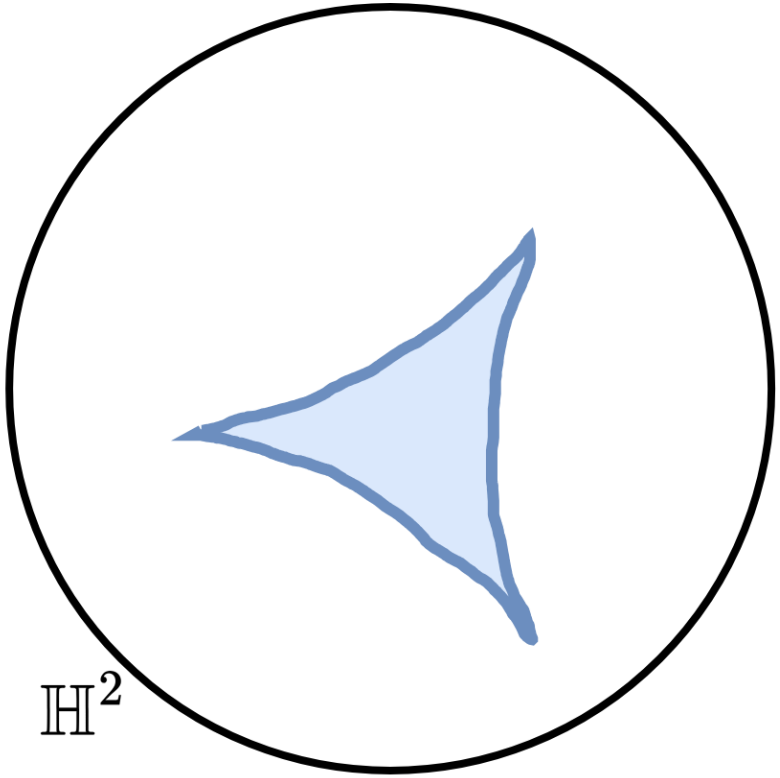
October 29, 2026

Department of Mathematics, Statistics, and Computer Science

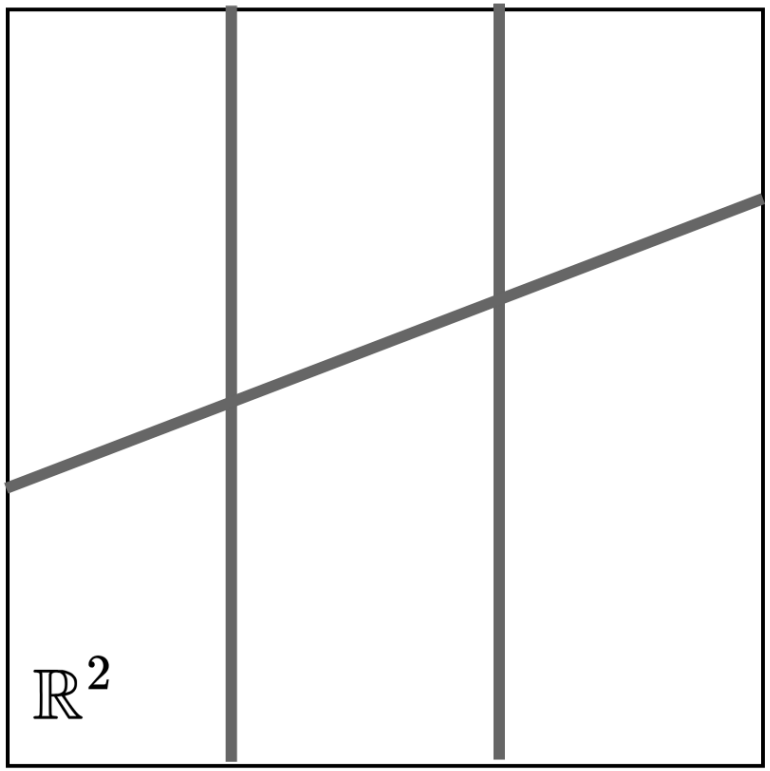
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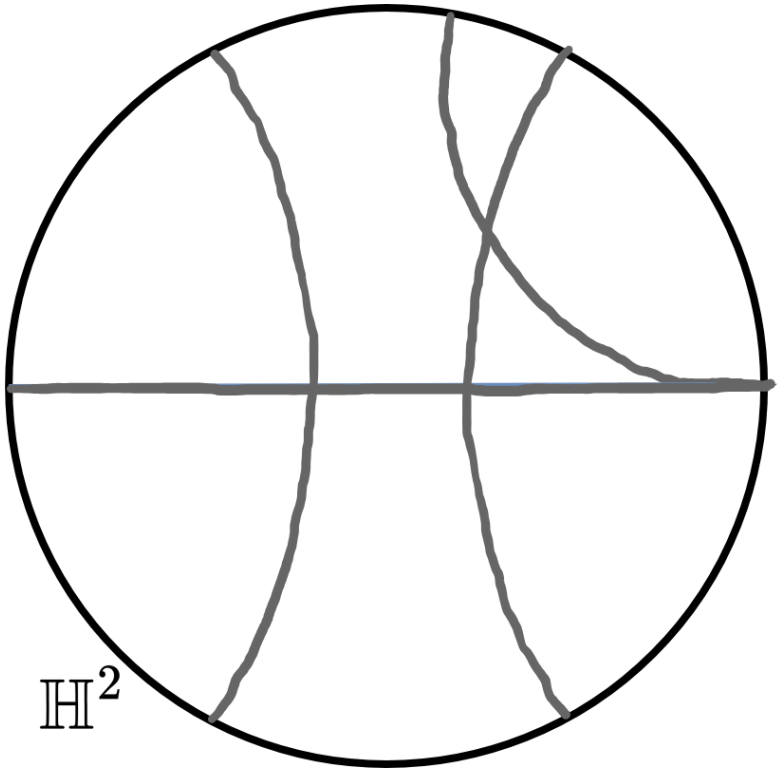
Hyperbolic dynamics



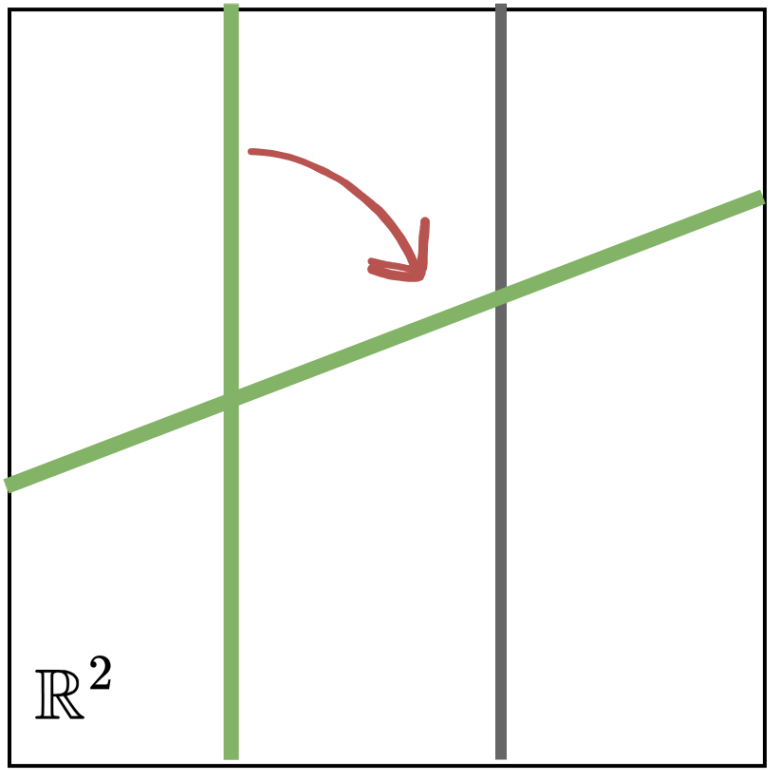


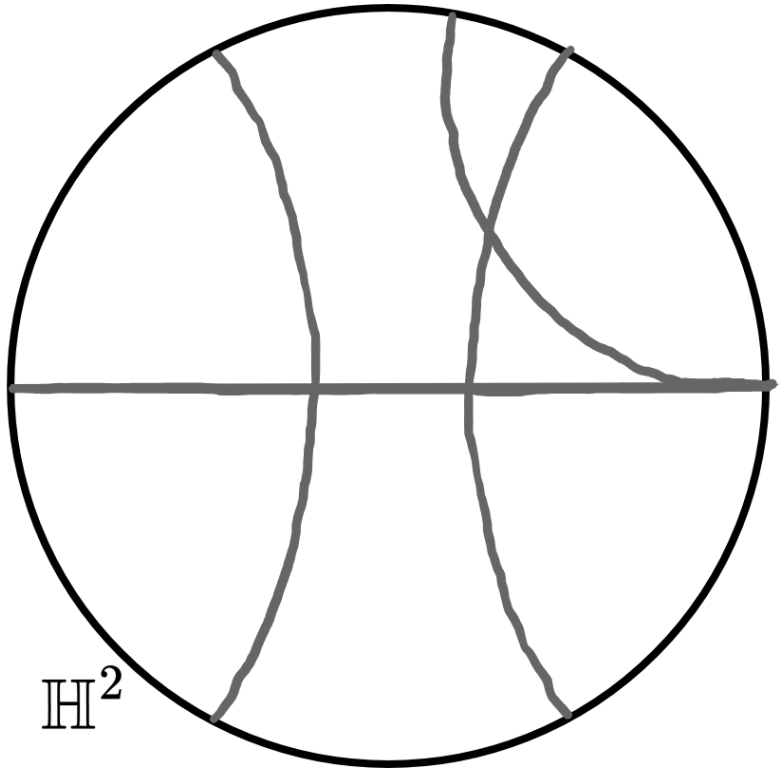
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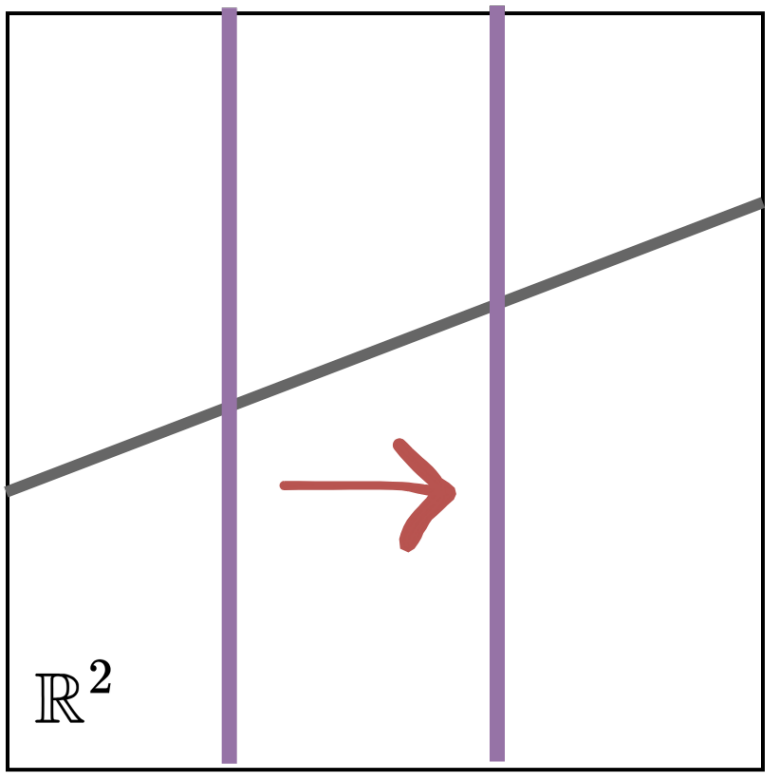


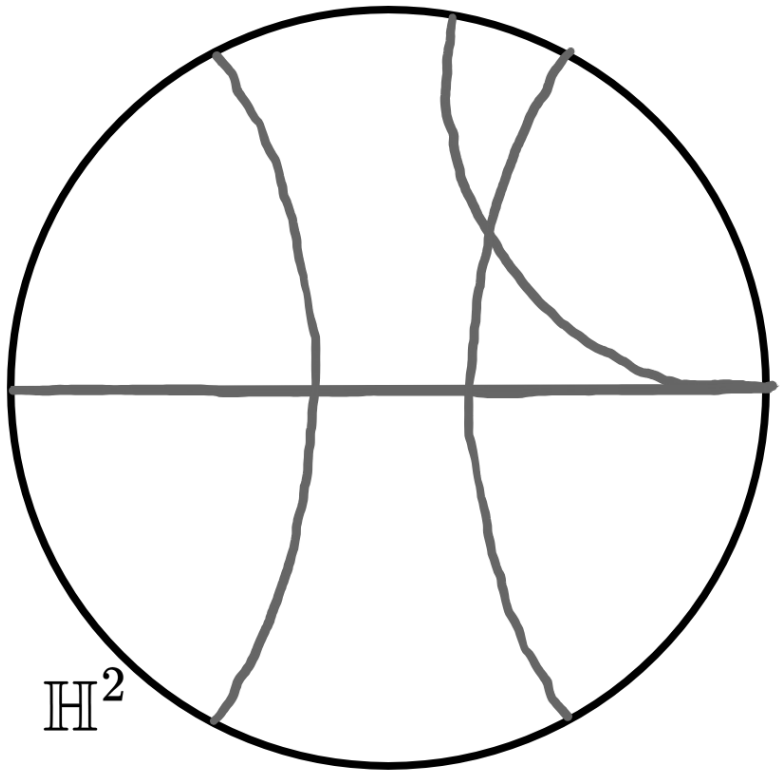
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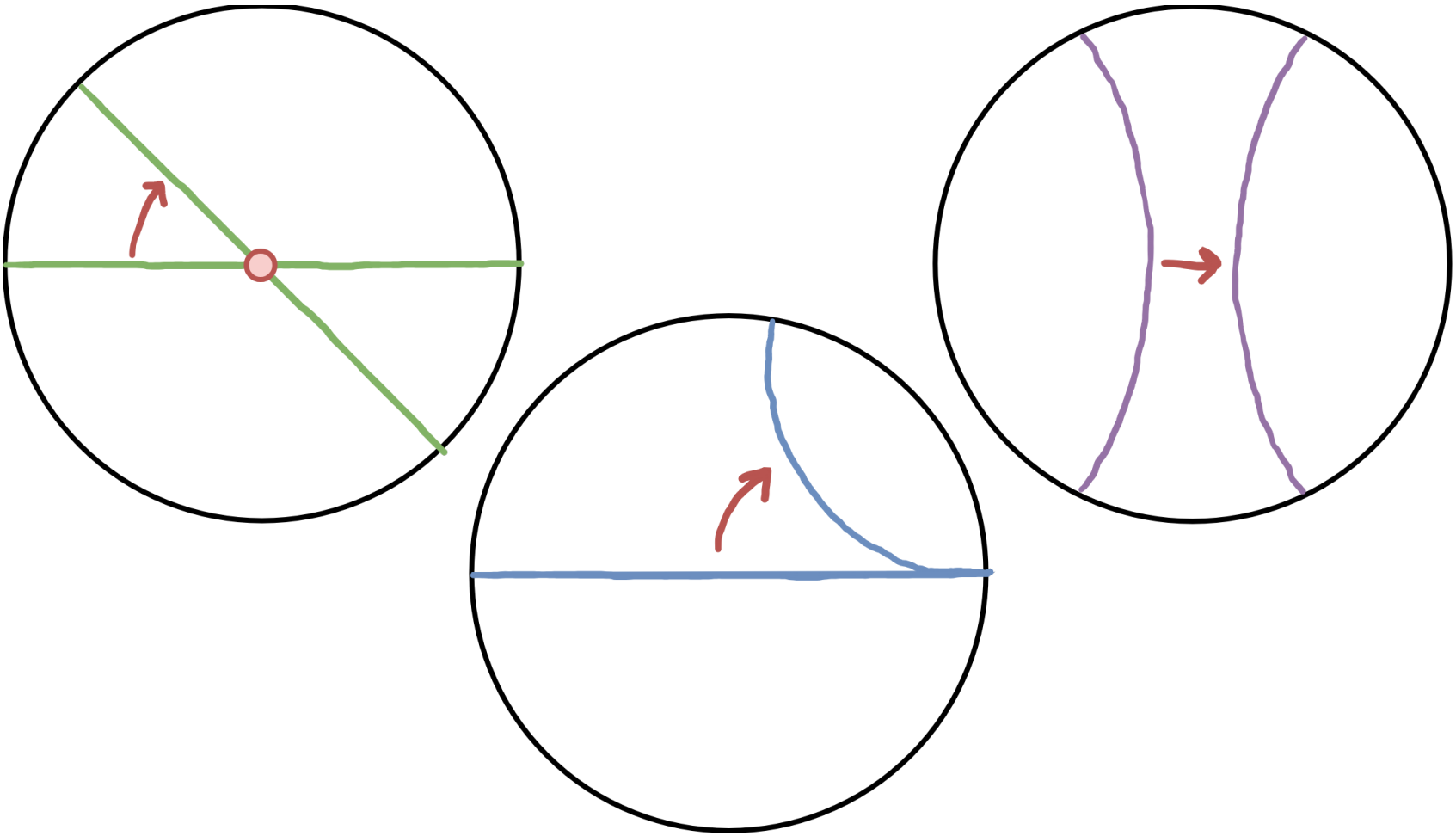


Hyperbolic dynamics

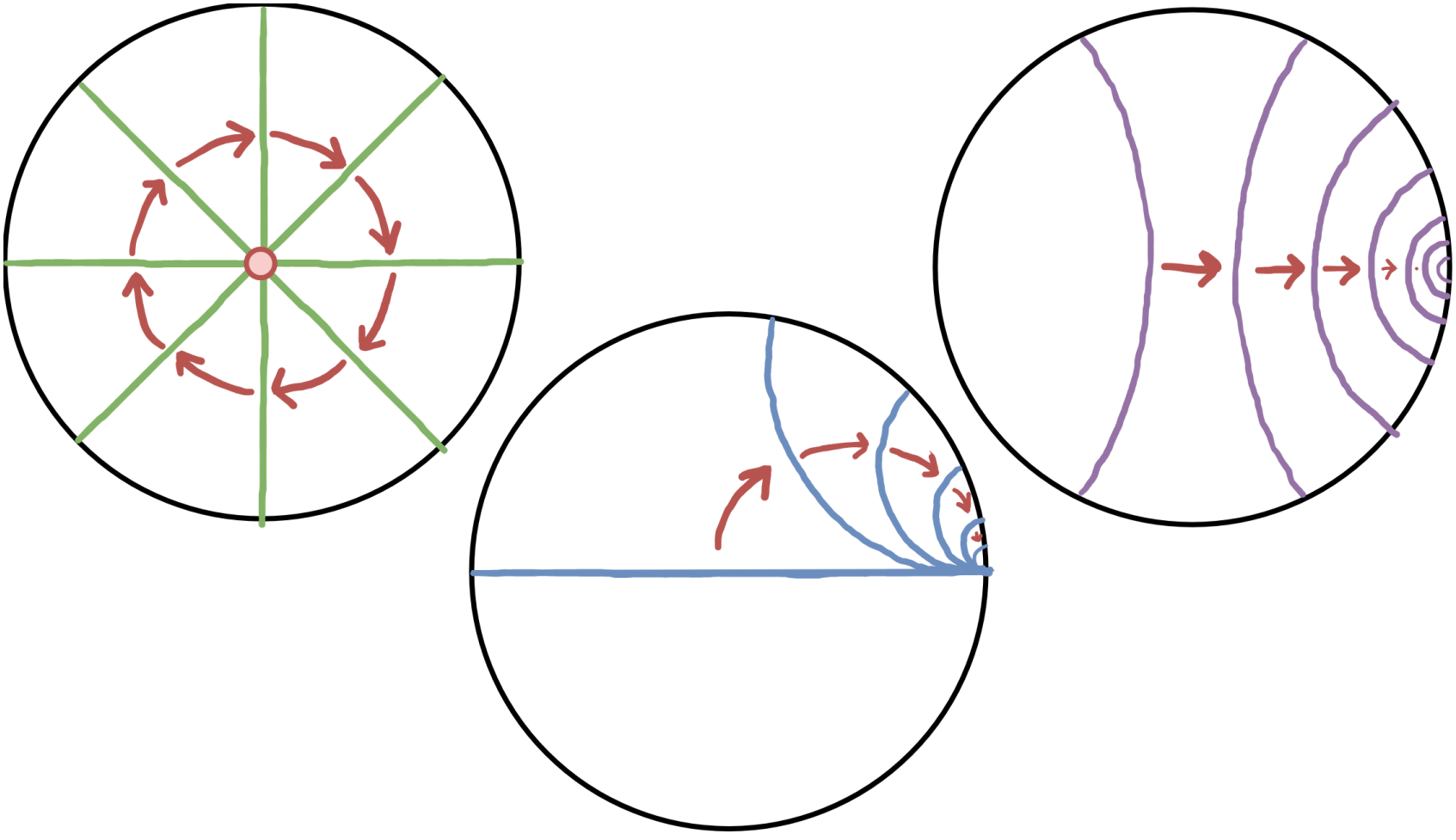




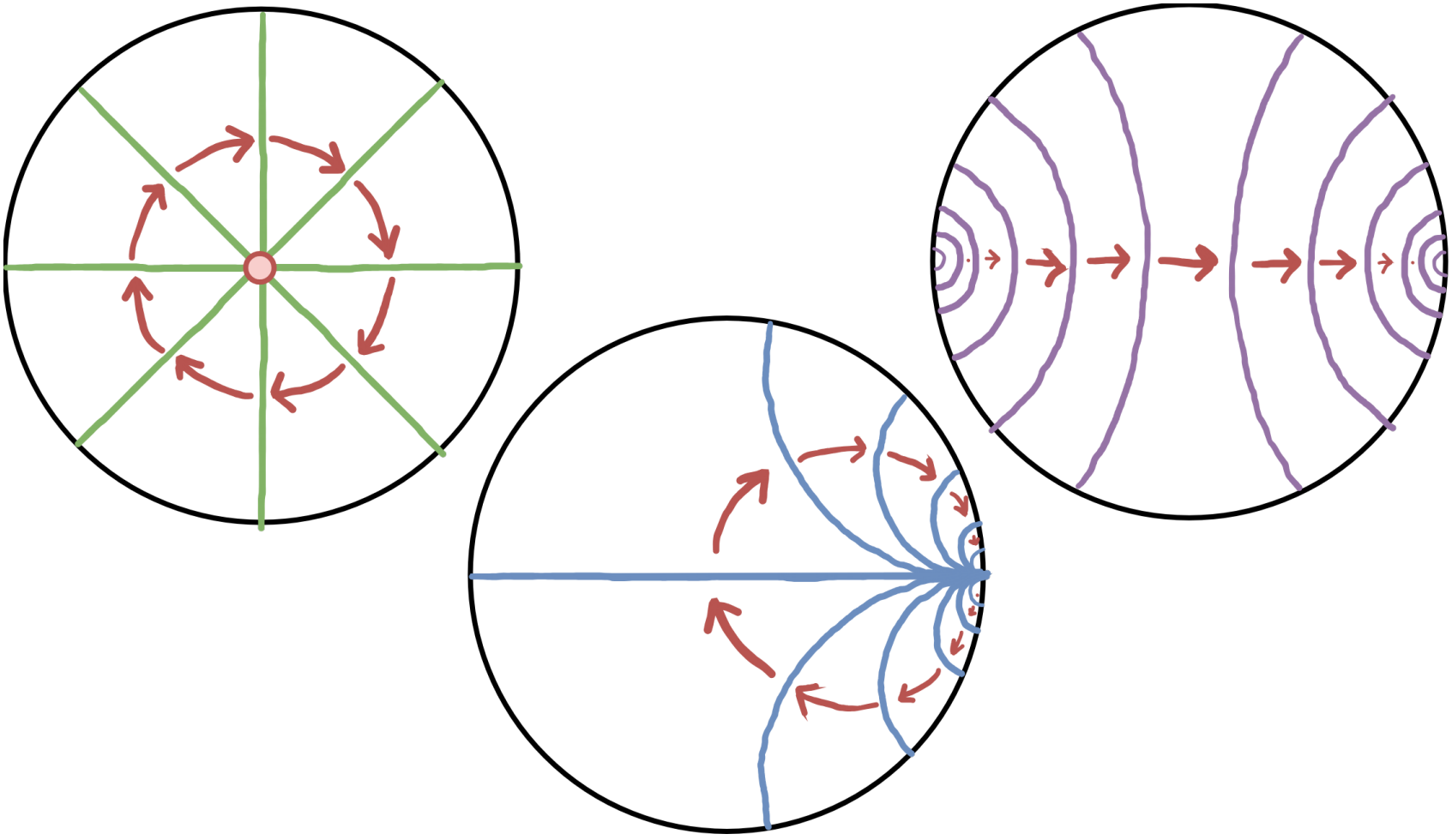
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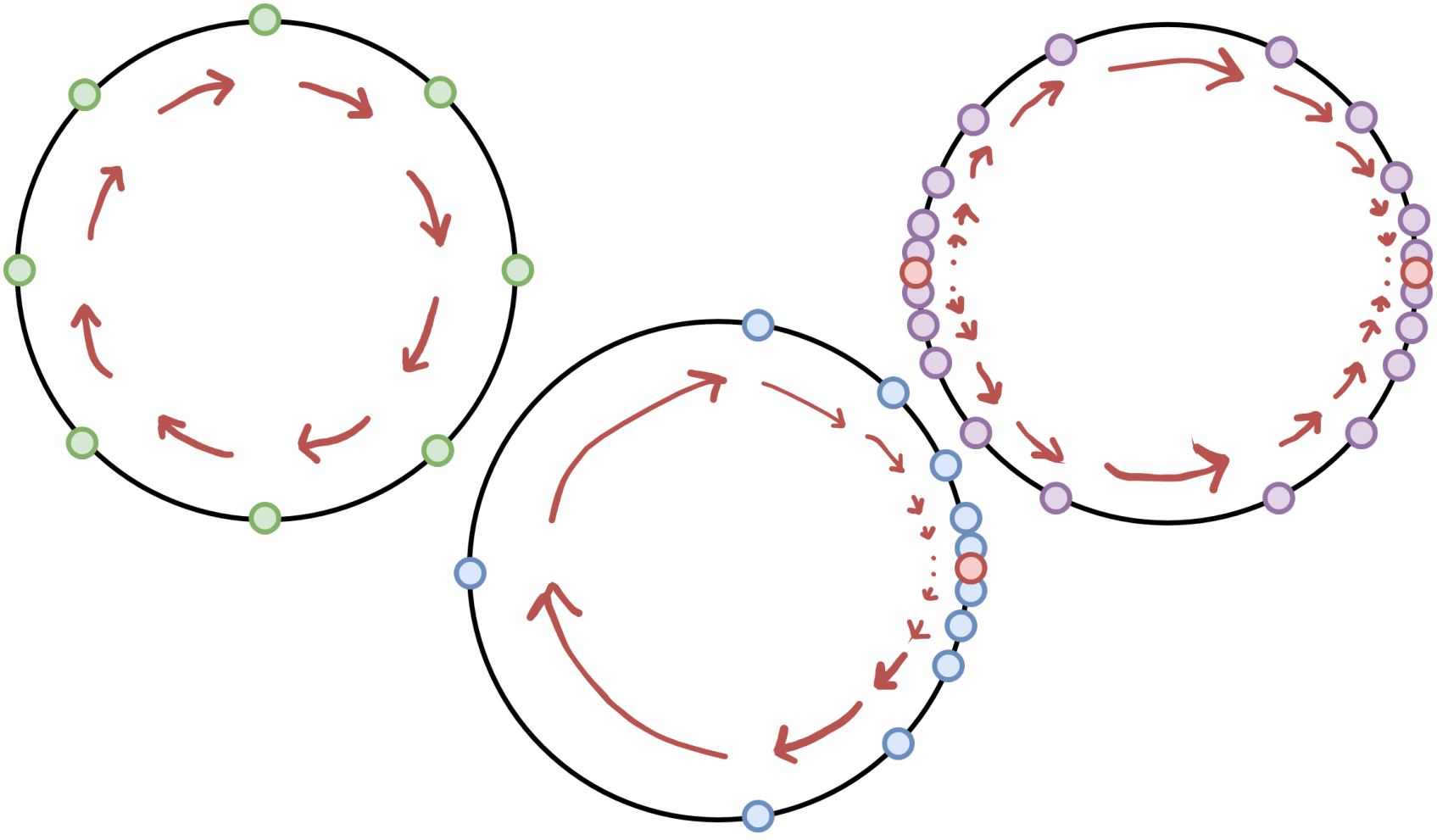
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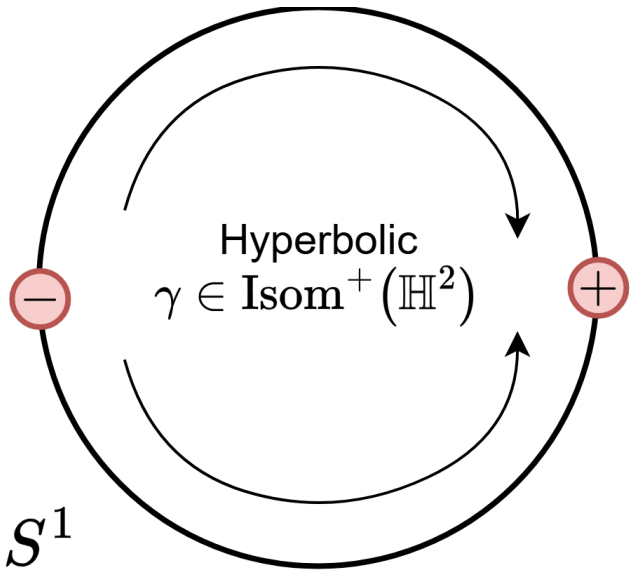
Hyperbolic dynamics



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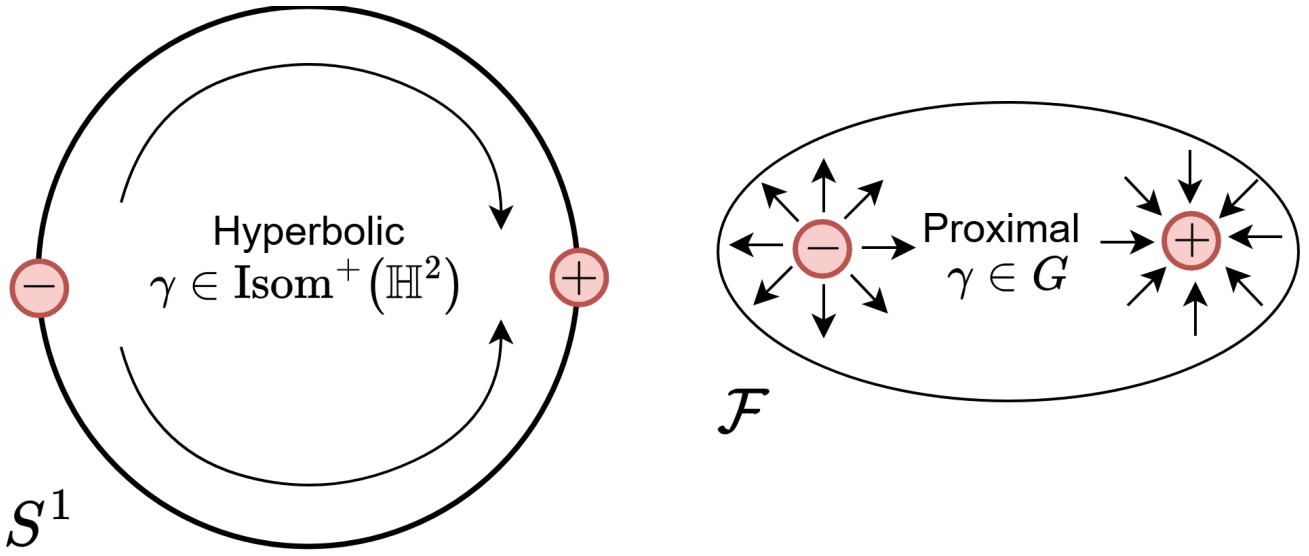


Hyperbolic dynamics



Dynamics of hyperbolic $\gamma \in \mathrm{Isom}^+(\mathbb{H}^2)$ on S^1.

Hyperbolic dynamics



Dynamics of hyperbolic $\gamma \in \mathrm{Isom}^+(\mathbb{H}^2)$ on S^1.

Dynamics of proximal $\gamma \in G$ on $\mathcal{F}$.

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Background: Flag varieties and Richardson varieties

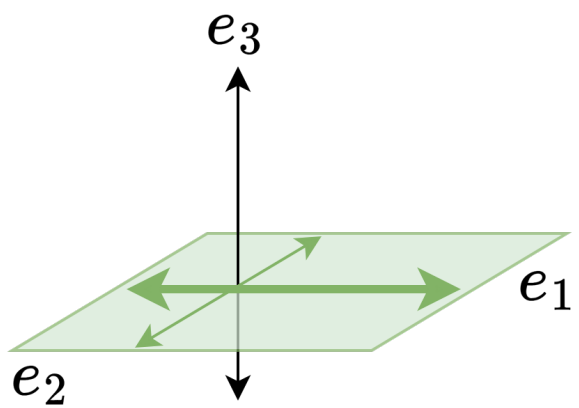
Flags in $\mathbb{C}^n$

Consider $\mathbb{C}^n$ with basis $(e_1, e_2, \ldots, e_n)$.

A *flag* is a tuple of nested subspaces in $\mathbb{C}^n$, e.g.

$$\text{Base flag} = (e_1, e_1 \oplus e_2, \ldots, e_1 \oplus \cdots \oplus e_{n-1}).$$

The *flag variety* $\mathcal{F}(\mathbb{C}^n)$ is the collection of such nested subspaces.



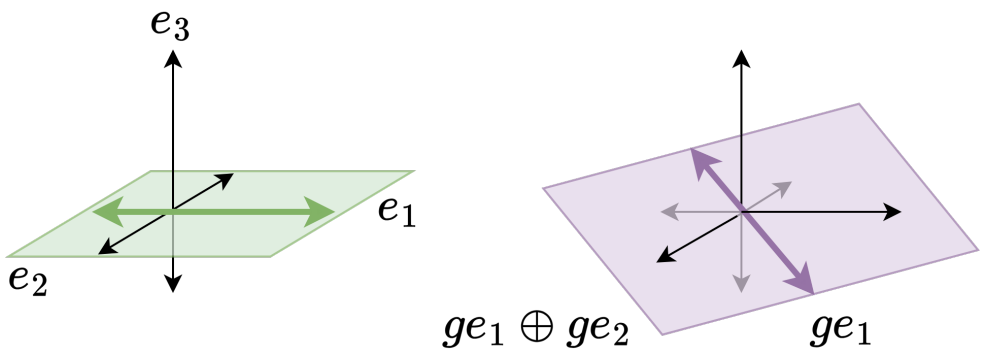
$$(e_1, e_1 \oplus e_2) \in \mathcal{F}(\mathbb{C}^3)$$

Action of $SL_n\mathbb{C}$ on $\mathcal{F}(\mathbb{C}^n)$

$SL_n\mathbb{C}$ acts on $\mathcal{F}(\mathbb{C}^n)$ by left multiplication.
The upper triangular matrices $B < SL_n\mathbb{C}$ stabilize the base flag.

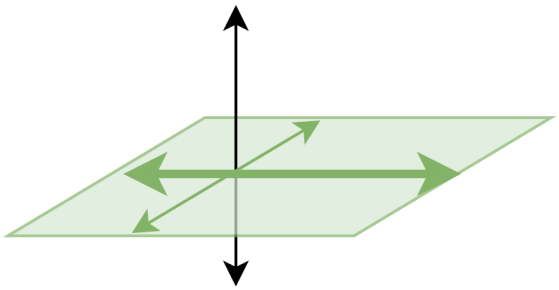
$$B = \begin{pmatrix} * & \cdots & * \\ & \ddots & \vdots \\ & & * \end{pmatrix} < SL_n\mathbb{C}$$

Thus $\mathcal{F}(\mathbb{C}^n) \cong SL_n\mathbb{C}/B$.



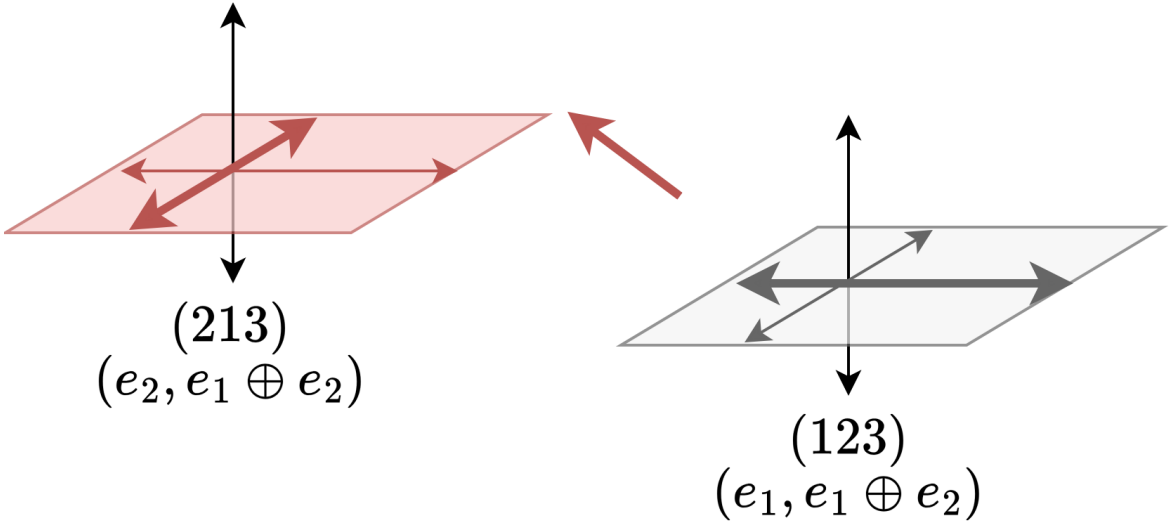
$eB = (e_1, e_1 \oplus e_2)$ $\qquad$ $gB = (ge_1, ge_1 \oplus ge_2)$

Action of permutations on $\mathcal{F}(\mathbb{C}^3)$

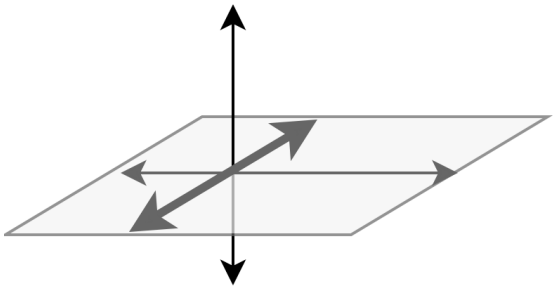


(123)
$(e_1, e_1 \oplus e_2)$

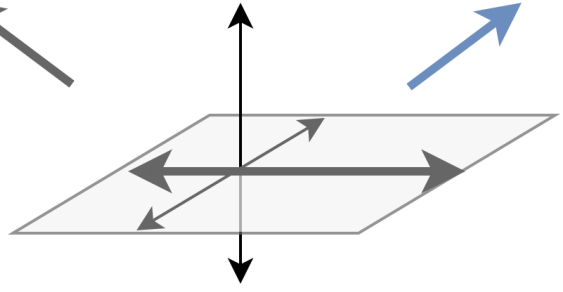
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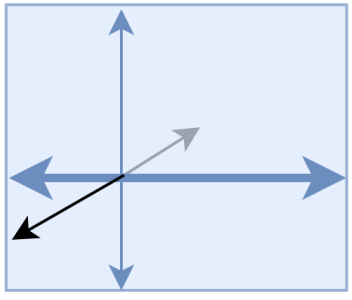
Action of permutations on $\mathcal{F}(\mathbb{C}^3)$



(213)
$(e_2, e_1 \oplus e_2)$

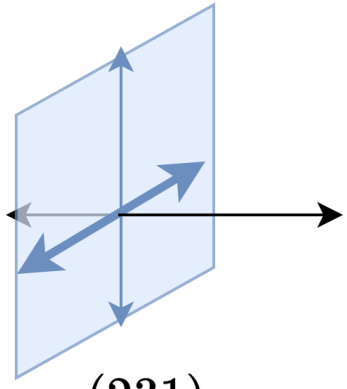


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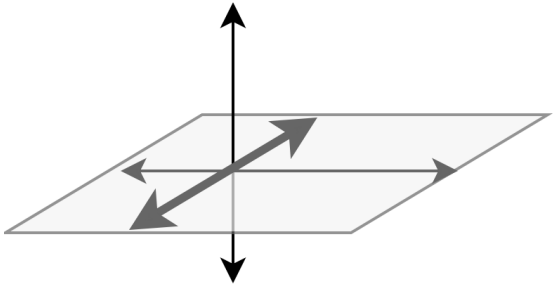


(132)
$(e_2, e_1 \oplus e_3)$

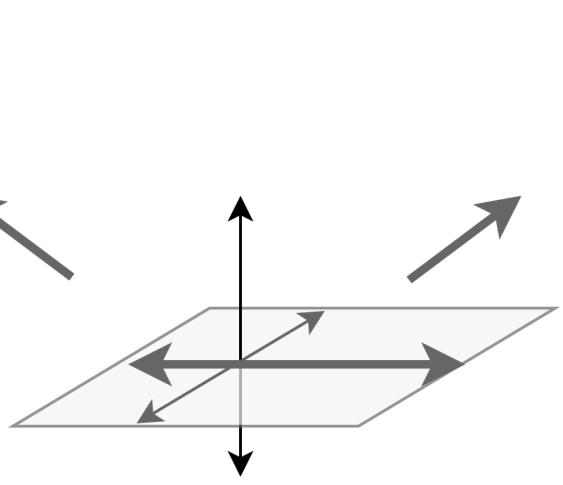
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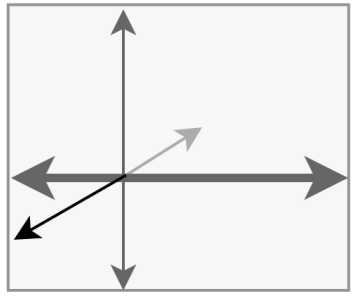
(231)
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(213)
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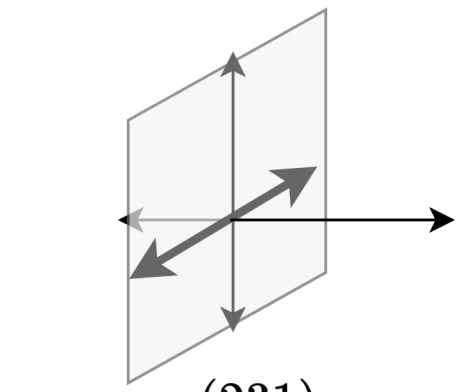


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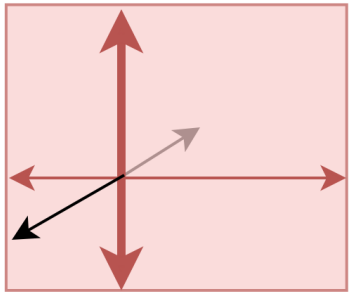


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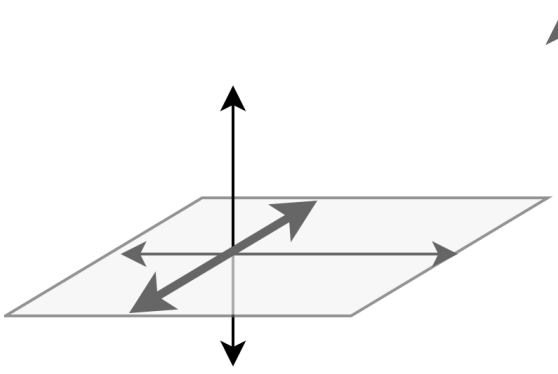
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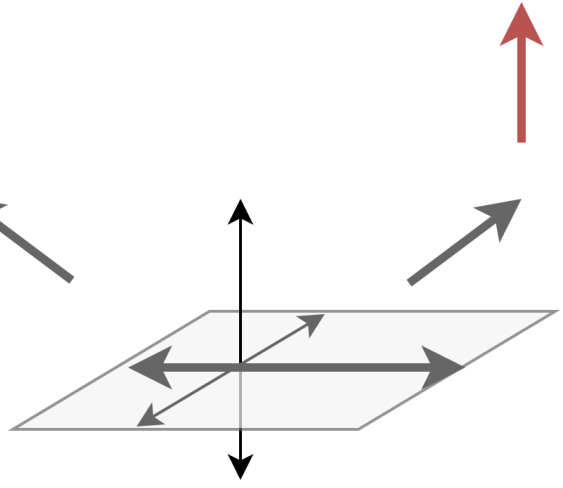
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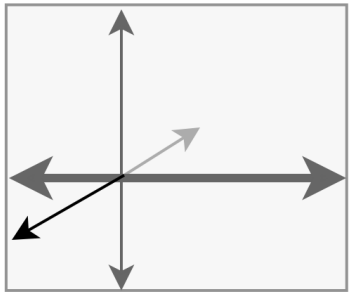
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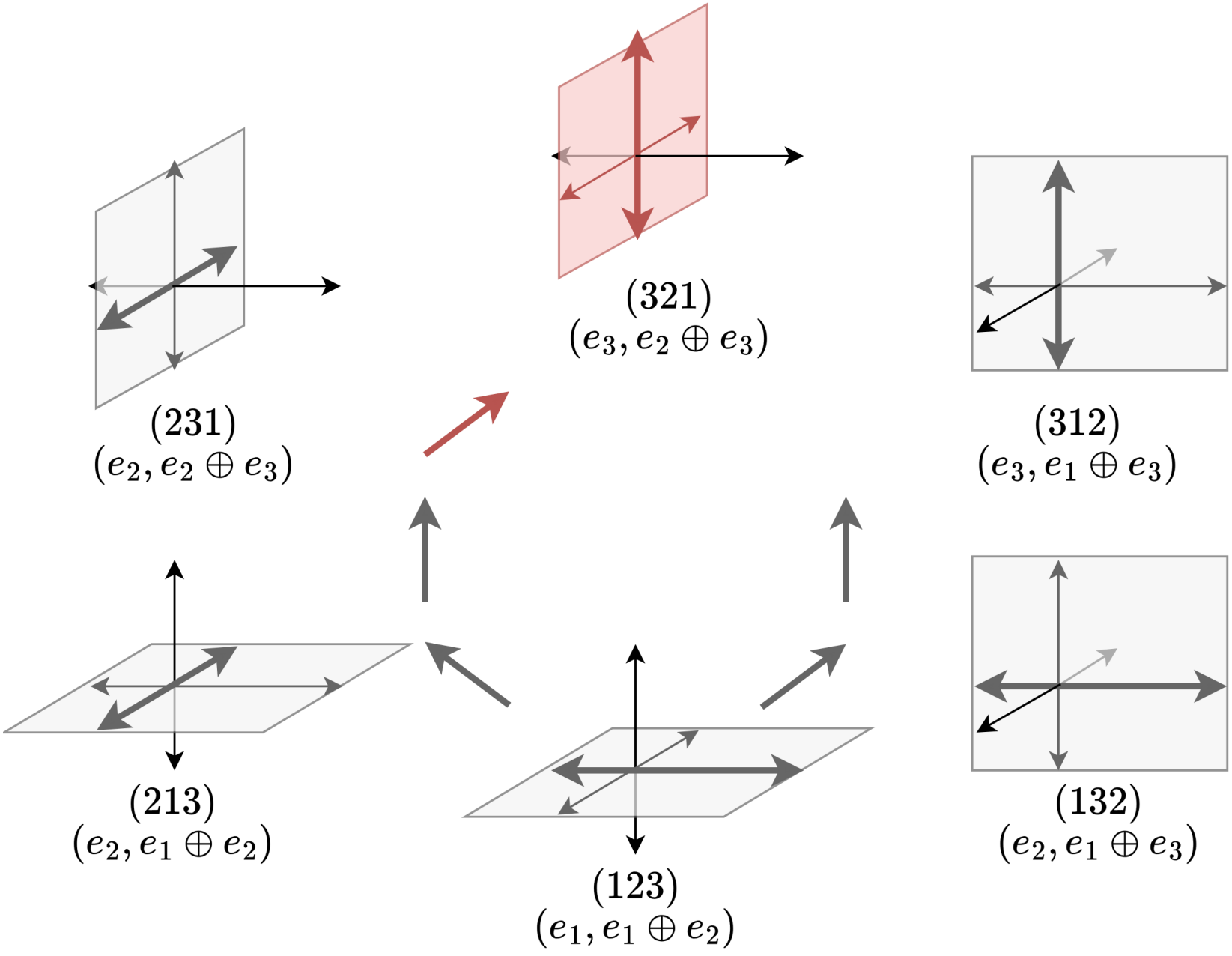


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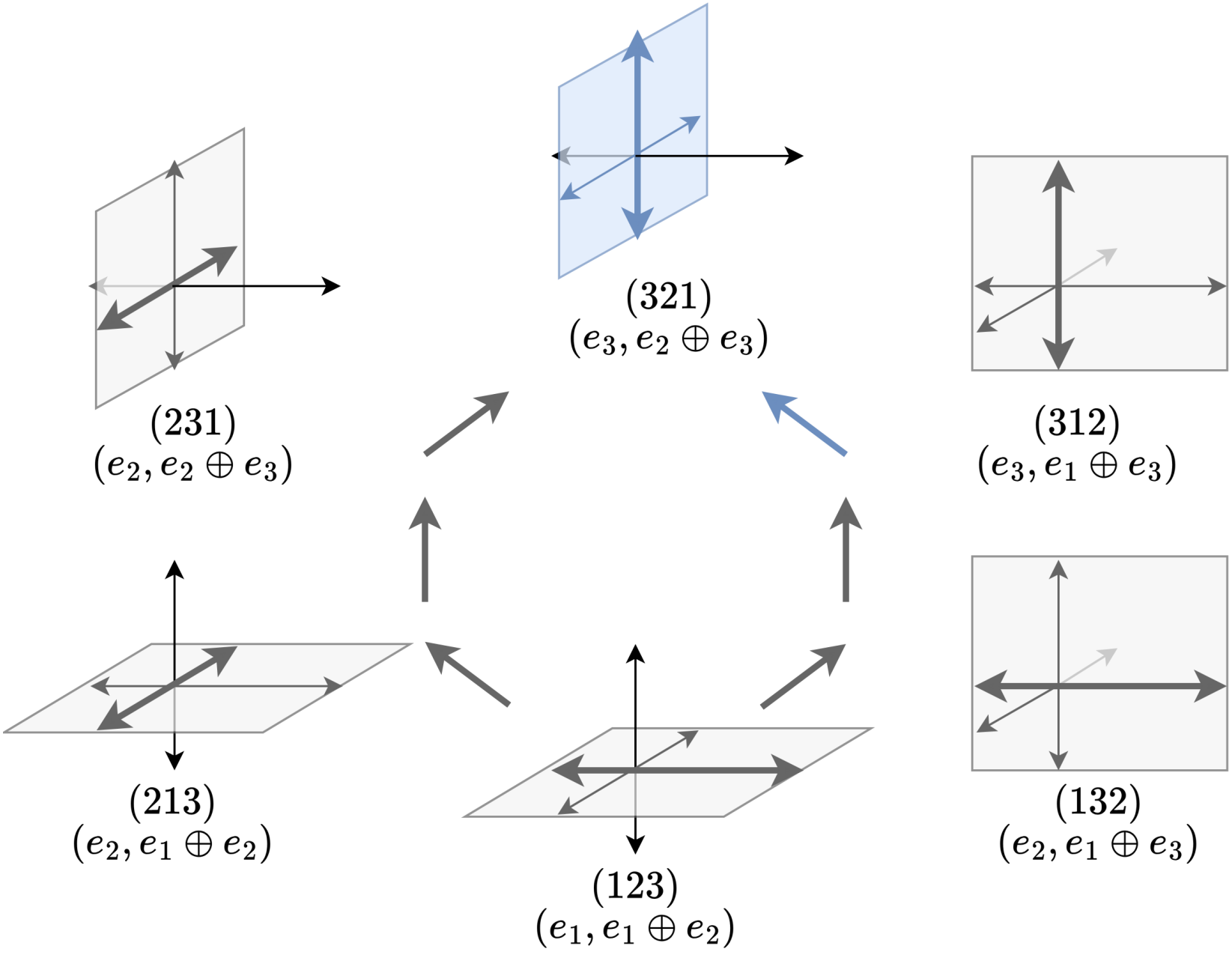


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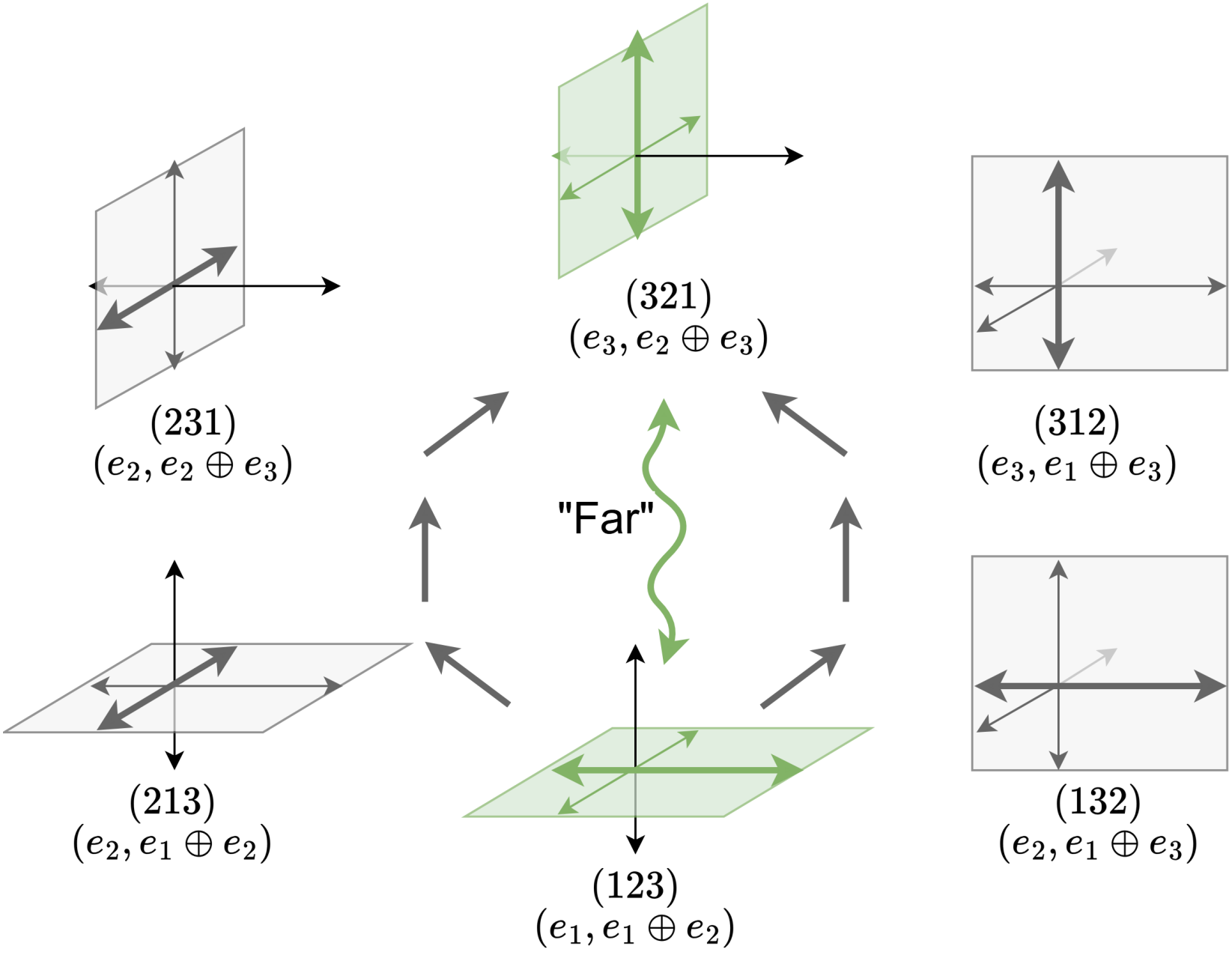
Action of permutations on $\mathcal{F}(\mathbb{C}^3)$



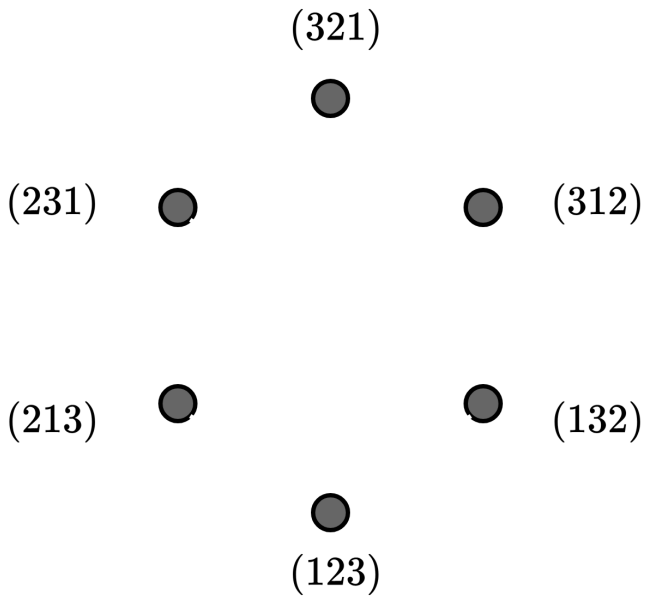
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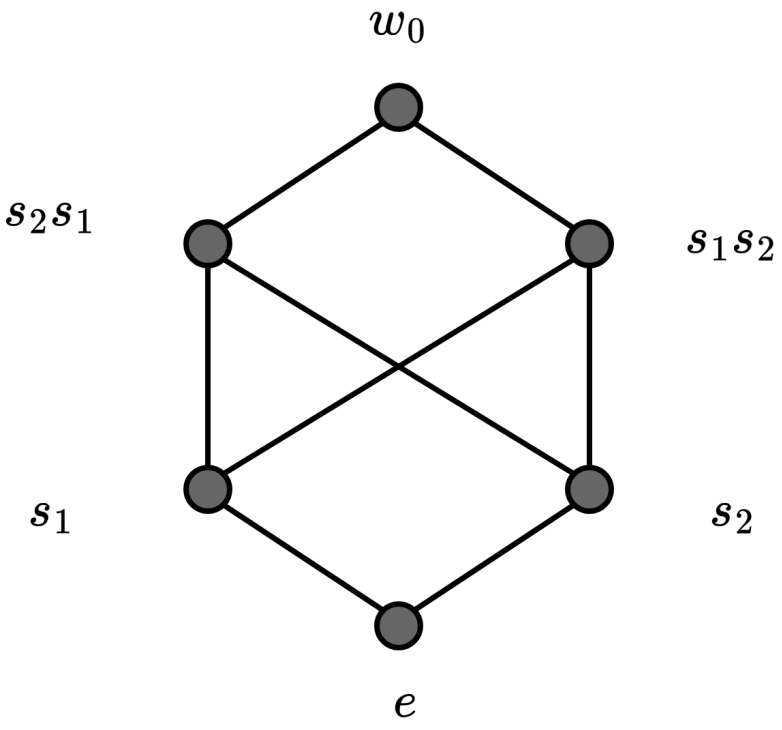


Weyl group of $\mathcal{F}(\mathbb{C}^3)$



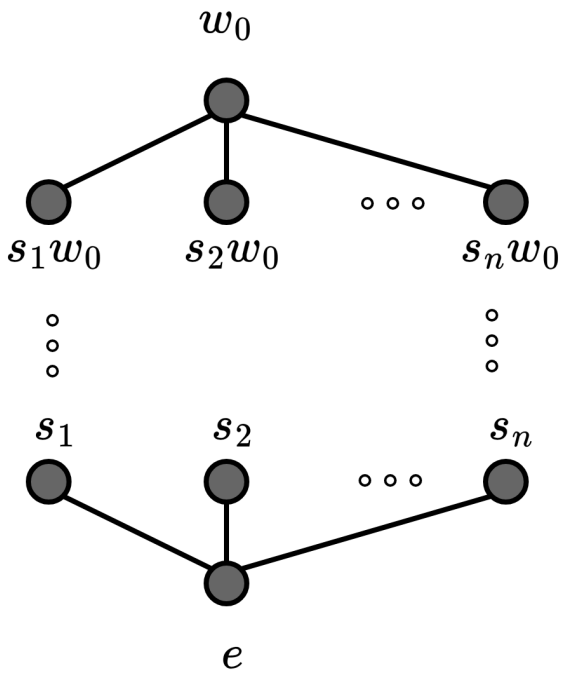
- $W = \mathrm{Sym}(3)$
- Generated by $s_1 = (1\ 2)$ and $s_2 = (2\ 3)$.
- $l(w)$ length of a minimal word
- Bruhat order $u \leq w$ minimal word containment
- Longest element w_0

Weyl group of $\mathcal{F}(\mathbb{C}^3)$



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Weyl group of $\mathcal{F}(\mathbb{C}^n)$



- $W = \mathrm{Sym}(n)$
- Generated by $s_i = (i \;\; i+1)$.
- $l(w)$ length of a minimal word
- Bruhat order $u \leq w$ minimal word containment
- Longest element w_0

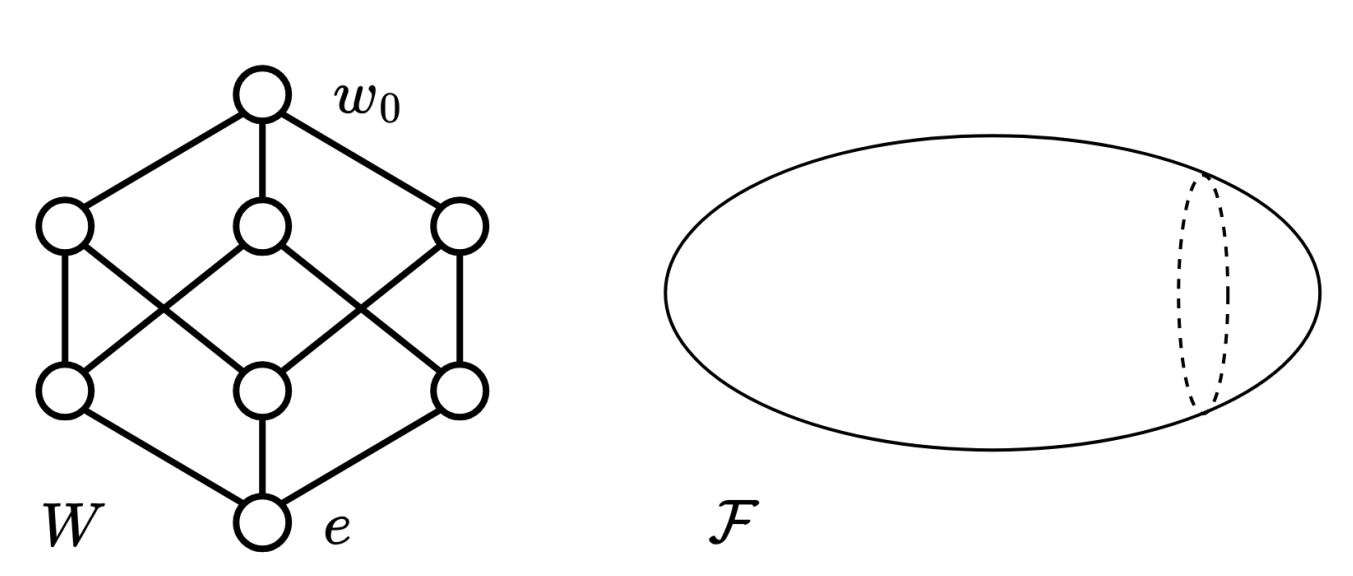
Generalizing the flag variety

$SL_n\mathbb{C}$	G	Complex semisimple Lie group
Upper triangular matrices	B	Minimal parabolic subgroup
$\mathcal{F}(\mathbb{C}^n)$	$\mathcal{F}$	G/B
$\mathrm{Sym}(n)$	W	Weyl group
	eB	
Flags	$gB, g \in G$	Flags
	$wB, w \in W$	

Schubert cells *BwB*

Let $w \in W$. We call $BwB \subset \mathcal{F}$ a *Schubert cell*.

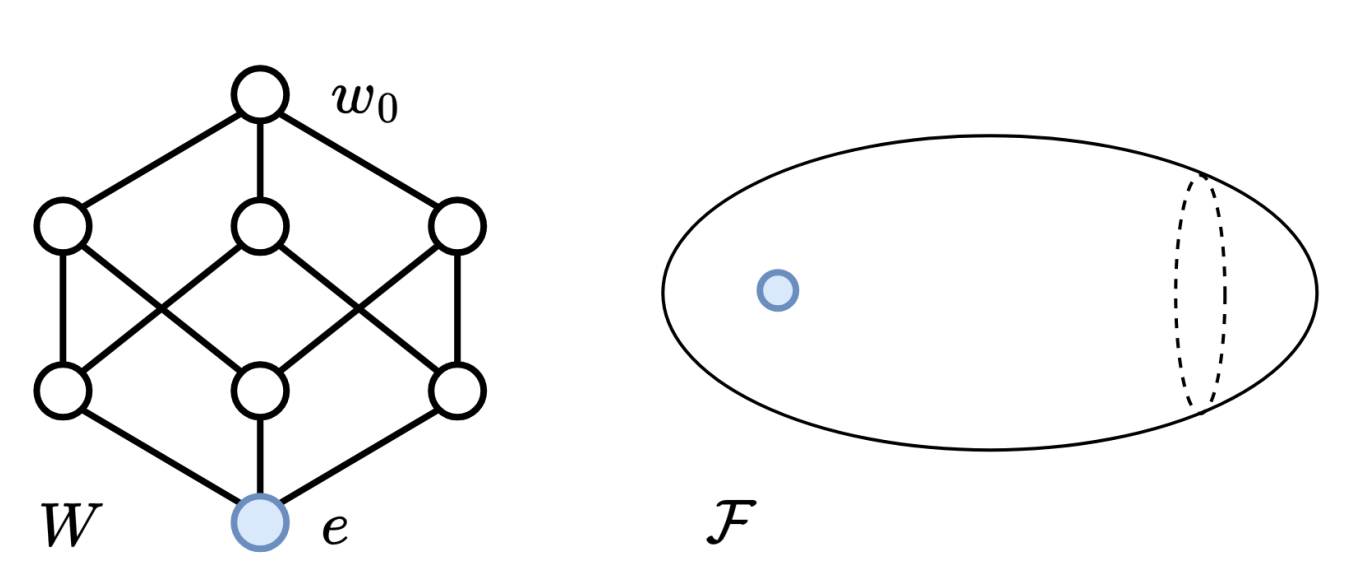
Then $BwB \cong \mathbb{C}^{l(w)}$.



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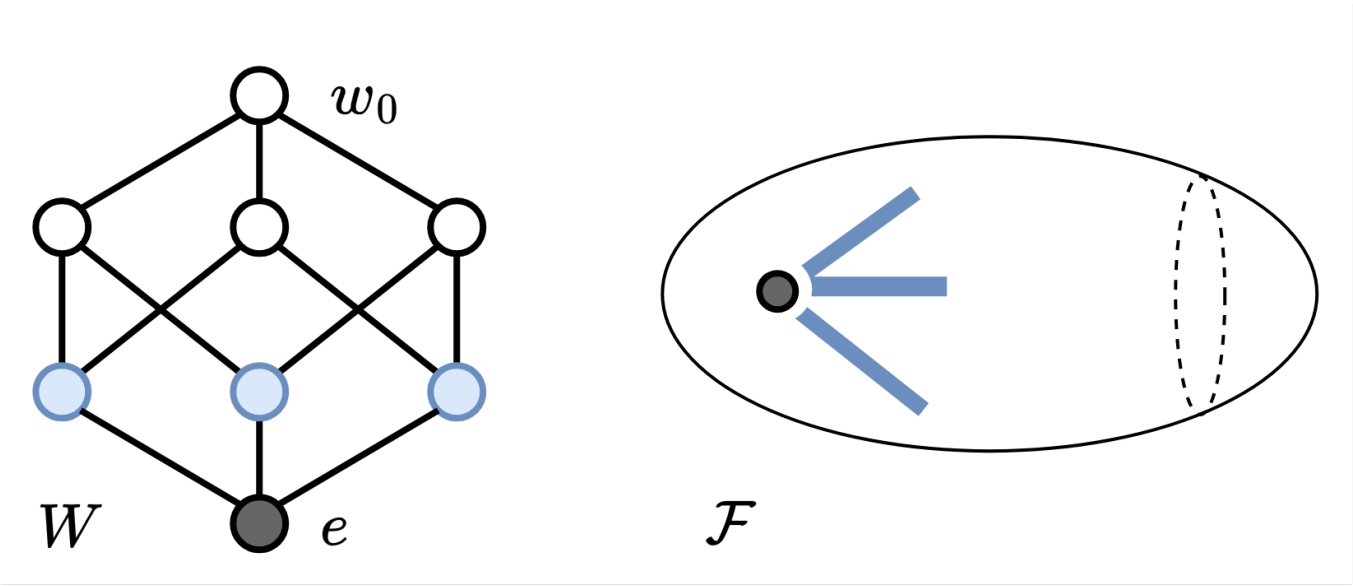
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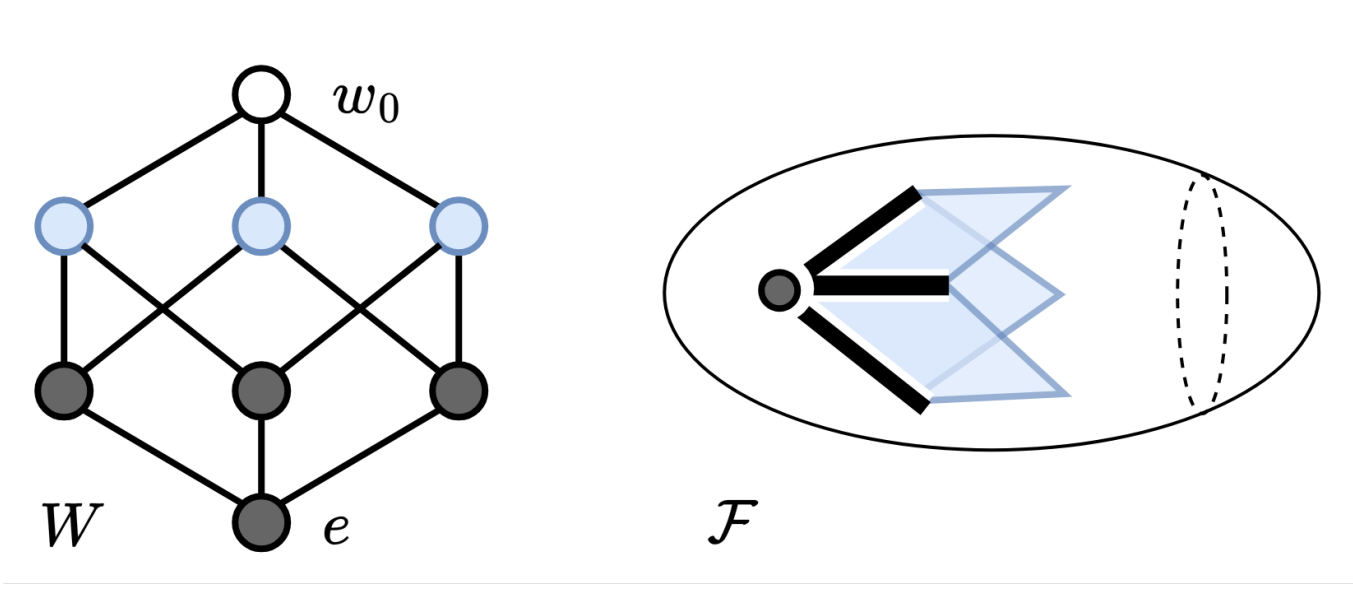
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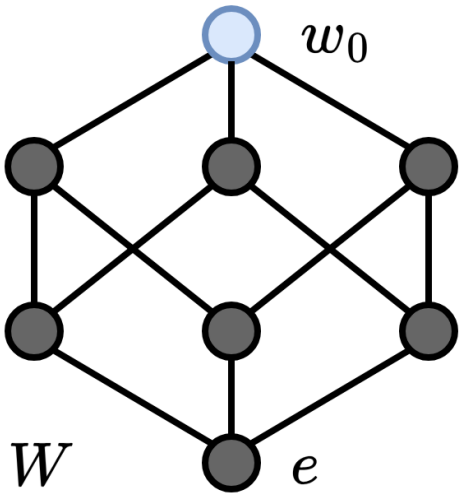
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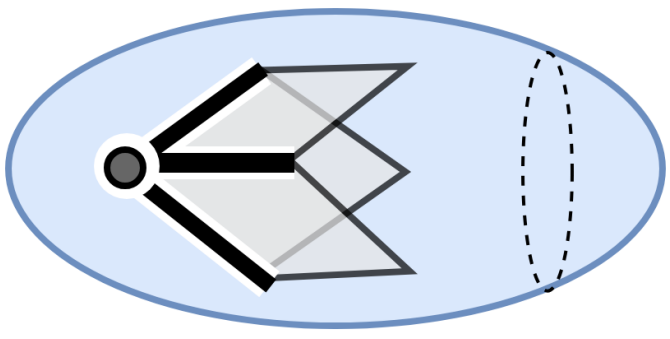


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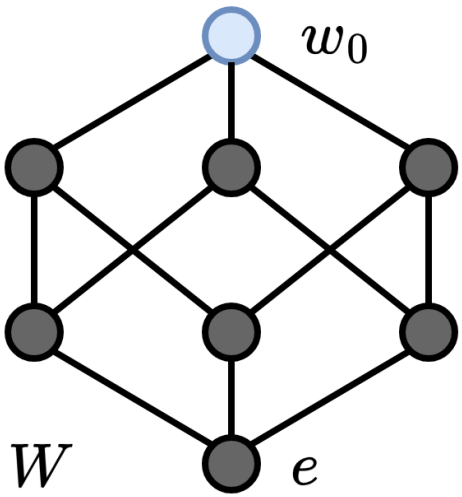
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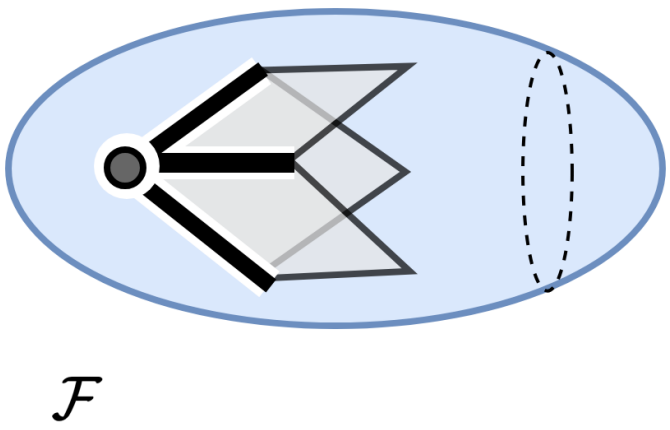
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The Schubert cells partition the flag variety (Bruhat, '56):

$$\mathcal{F} = \bigsqcup_{w \in W} BwB.$$



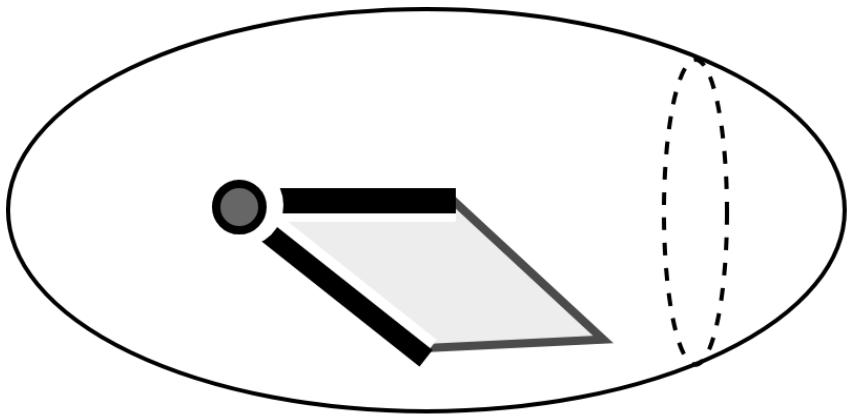


Schubert varieties $\alpha_w = \overline{BwB}$

Schubert varieties are the closure of Schubert cells:

$$\alpha_w = \overline{BwB}$$

$$\alpha_w = \{y \in \mathcal{F} : w(y) \leq w\}$$

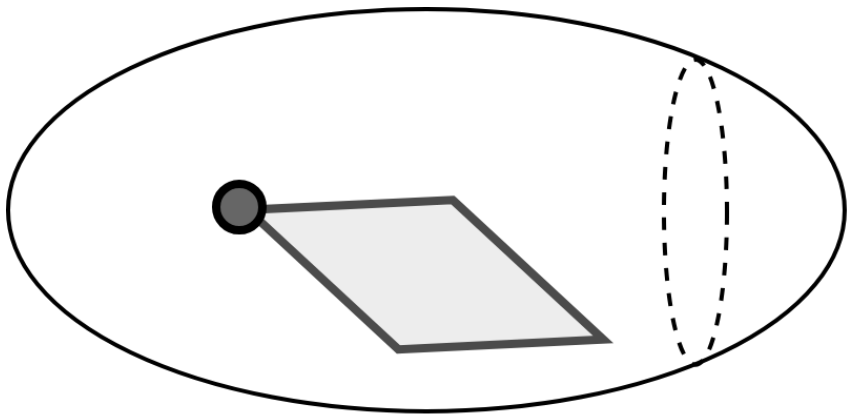


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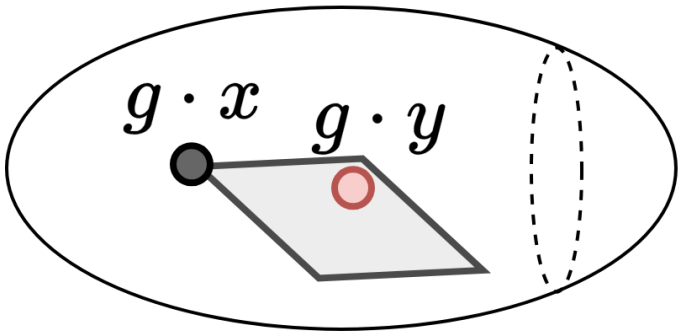


Schubert varieties $\alpha_w(x)$

A Schubert variety based at $x \in \mathcal{F}$ is a translate of α_w.

$$\alpha_w(x) = \{y \in \mathcal{F} : w(x, y) \leq w\}$$

If $w(x, y) = w_0$, we call x and y *transverse*.

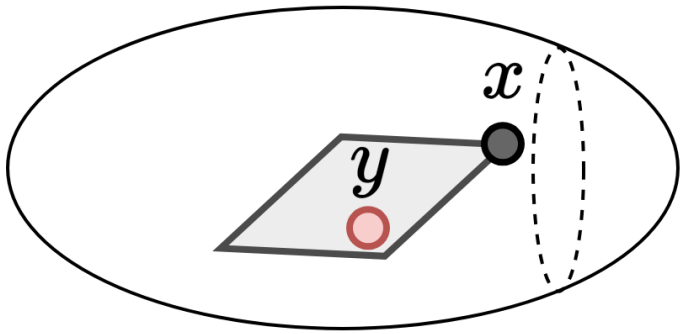


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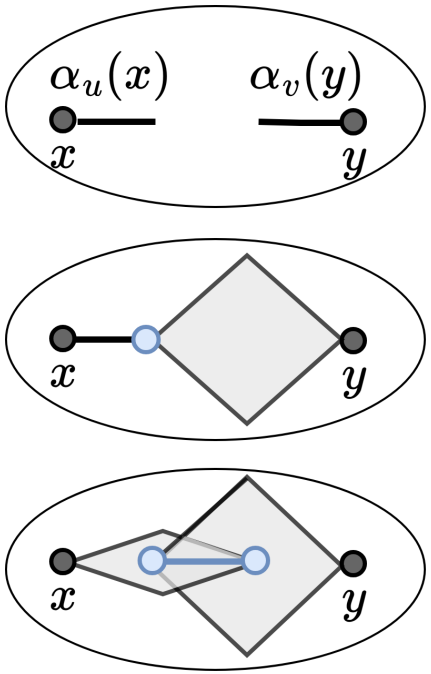
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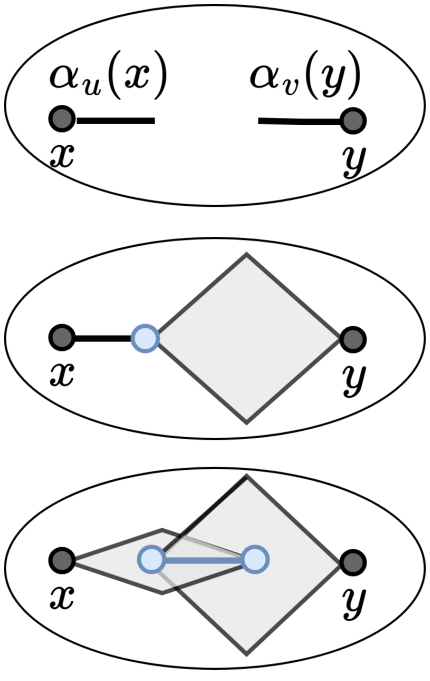
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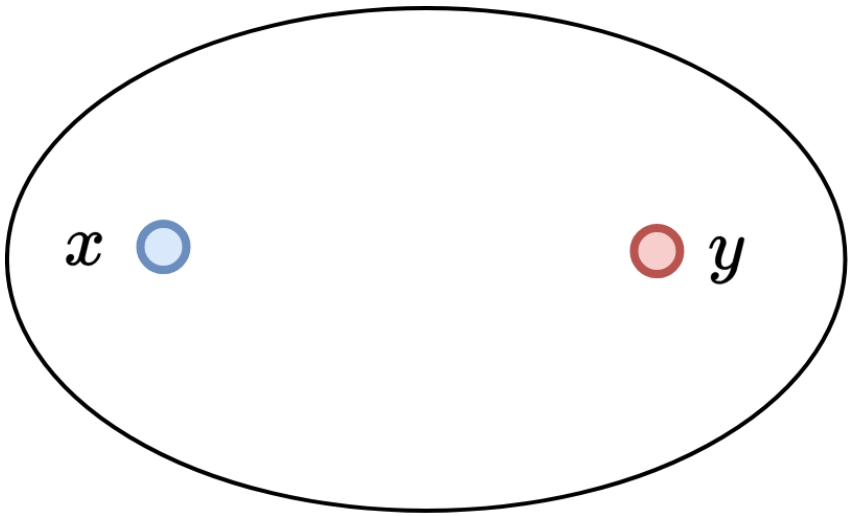
Dimension depends only on $u, v \in W$. Let $\iota = ww_0$.

- Empty if $\iota(v) \not\leq u$.
- Point if $\iota(v) = u$.
- Dimension $l(u) + l(v) - l(w_0)$ if $\iota(v) < u$.

Richardson 0-varieties

The 0-dimensional Richardson varieties $\mathrm{Rich}^0_{x,y}$ are in bijection with the Weyl group W:

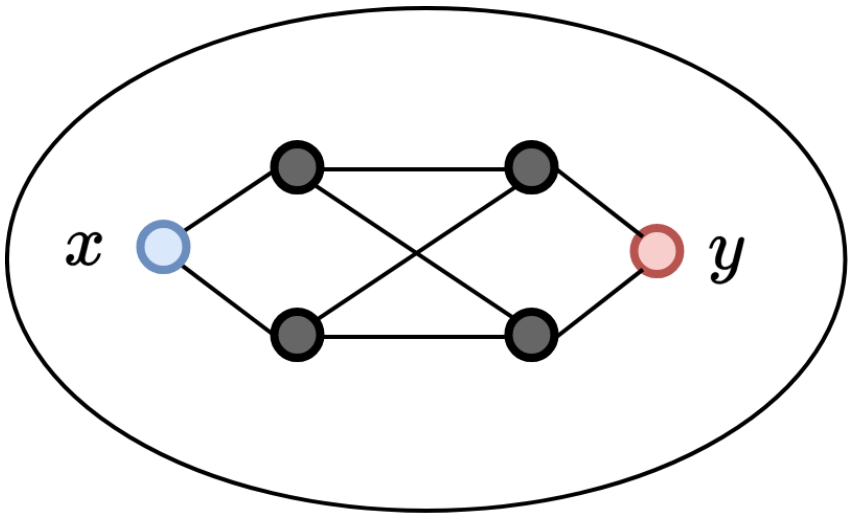
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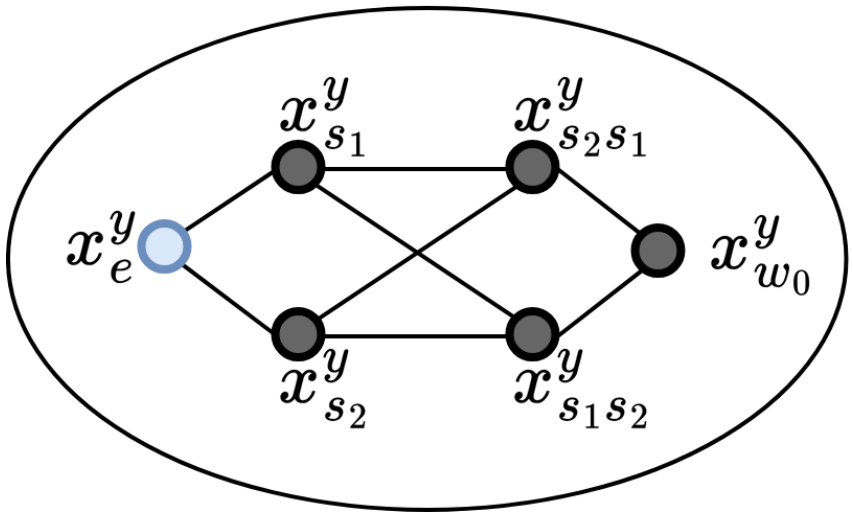
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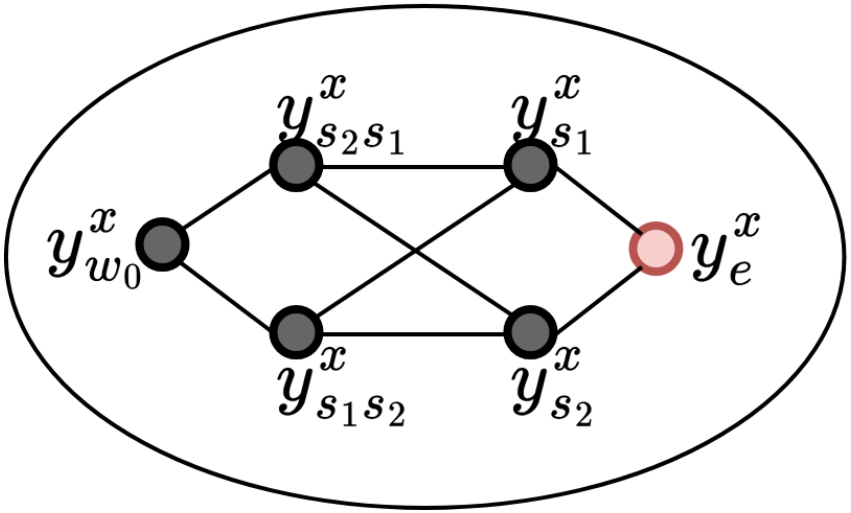
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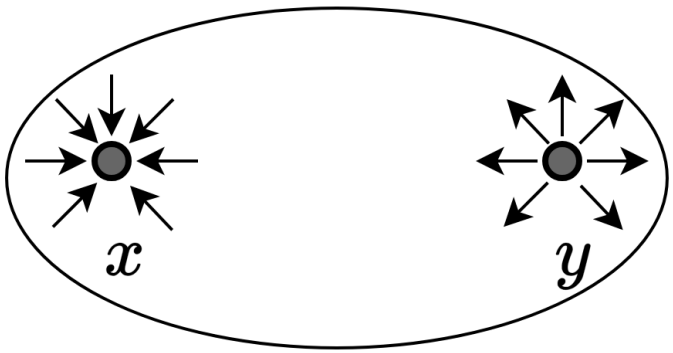
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Theorem A: Richardson varieties describe proximal dynamics of $\mathcal{F}$

Theorem A: Proximal dynamics on $\mathcal{F}$

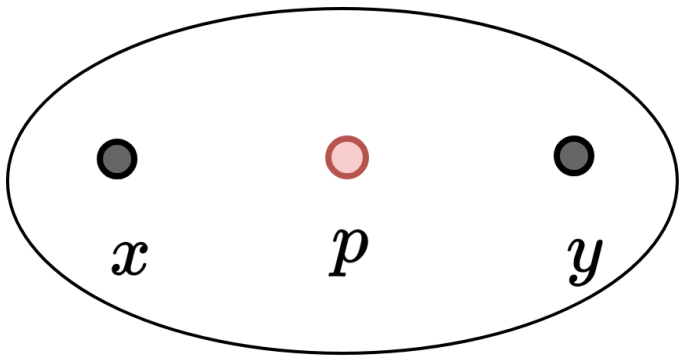
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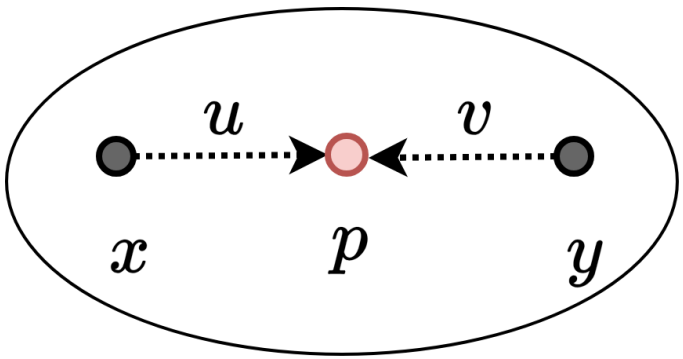
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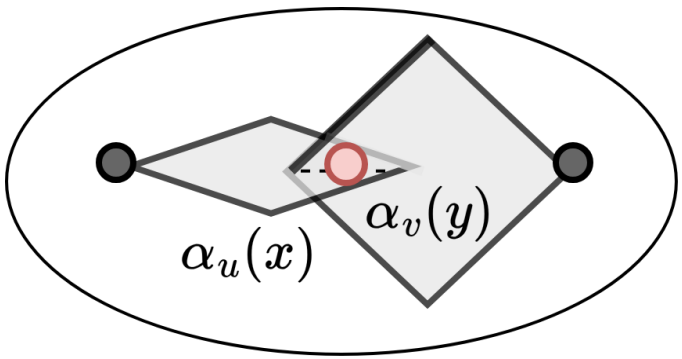
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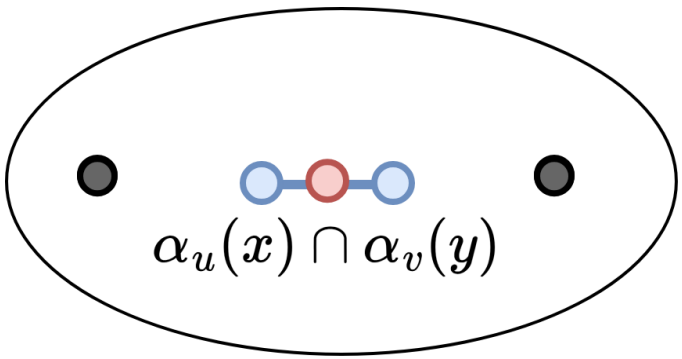
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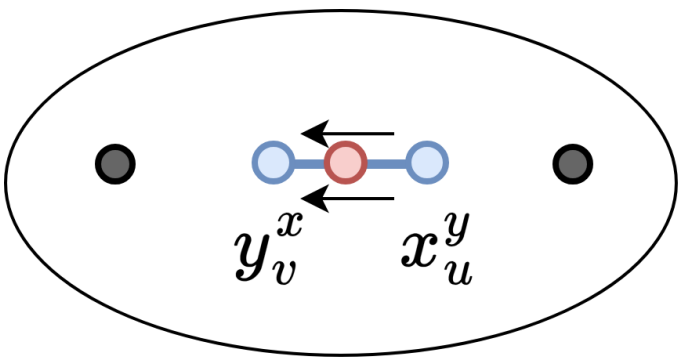
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Theorem A (V Chapter 3): Then

$$\lim_{i \to \infty} \gamma^i \cdot p = y_v^x,$$
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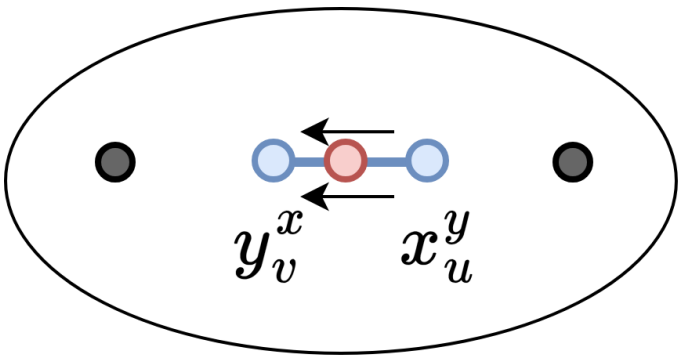
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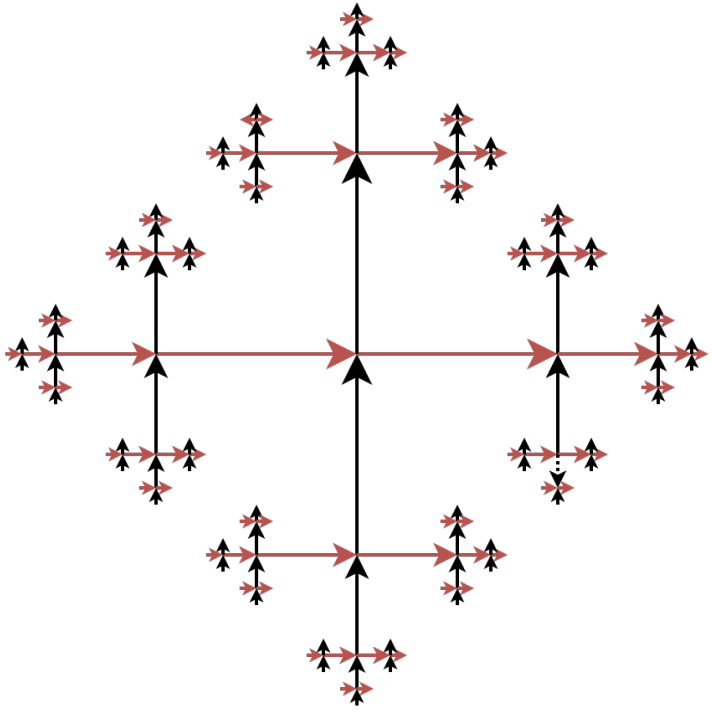
$$\lim_{i \to \infty} \gamma^i \cdot p = y^x_v,$$
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Corollary A: The fixed points of γ are $\mathrm{Rich}^0_{x,y}$.



Representations of hyperbolic groups

Hyperbolic groups



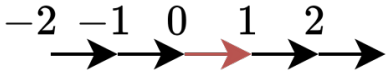


$F = \langle a, b \rangle$

Free group

$\pi_1 \Sigma = \langle a, b, c, d \mid [a, b][c, d] \rangle$

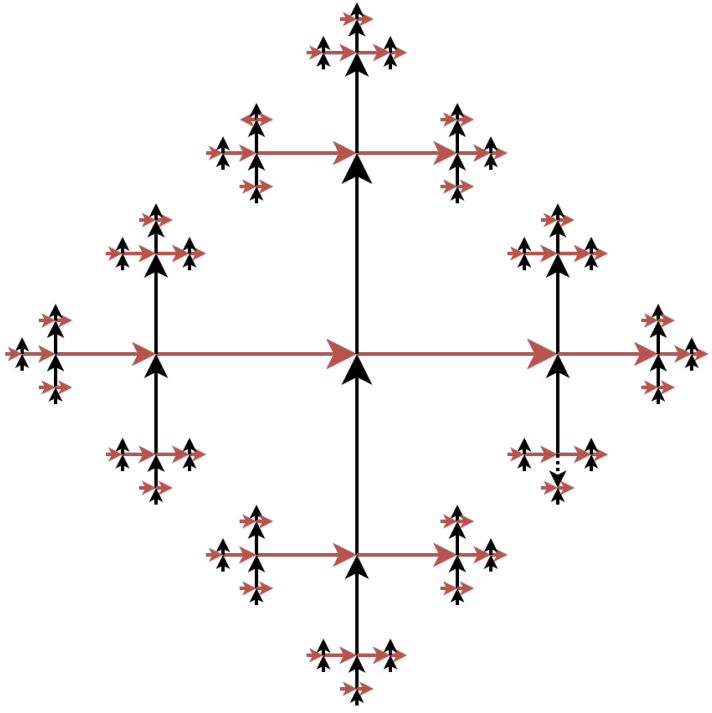
Surface group



$\mathbb{Z} = \langle a \rangle$

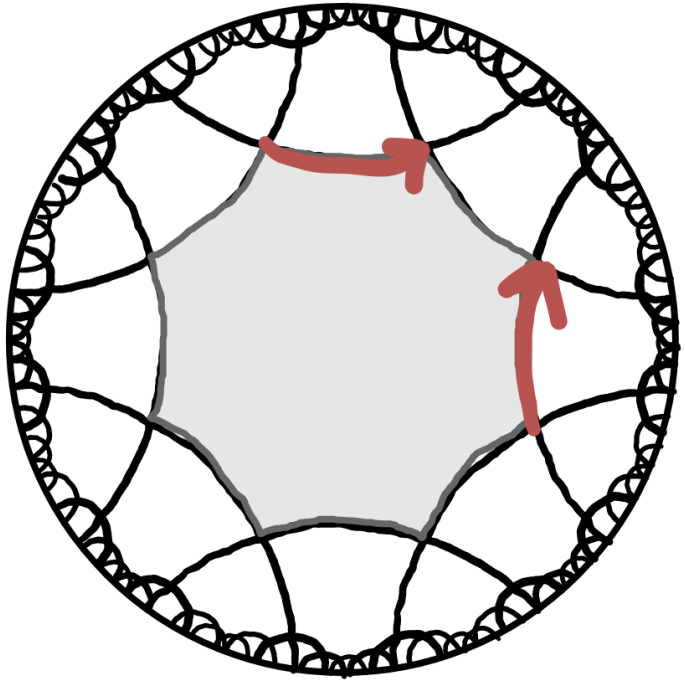
Integers

Hyperbolic groups



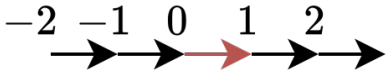
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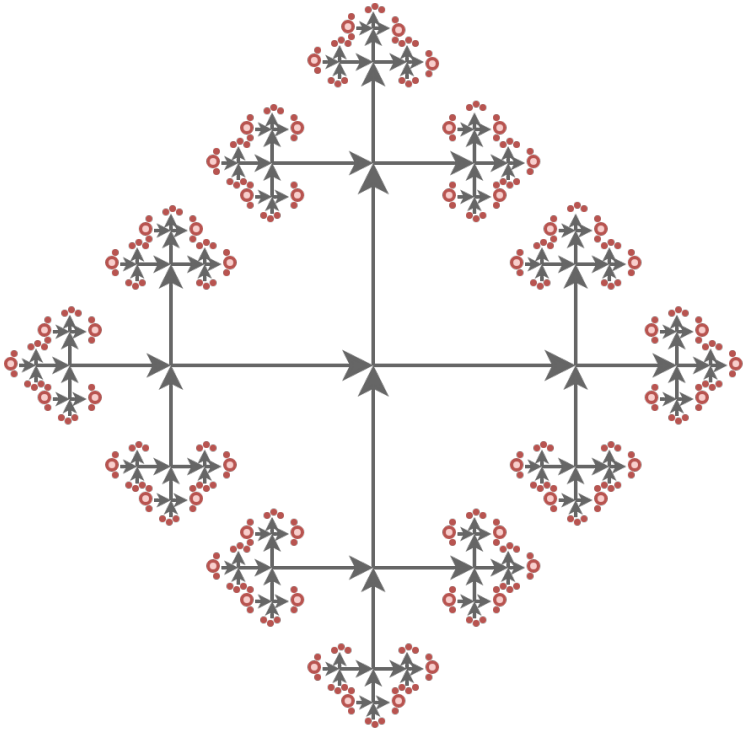
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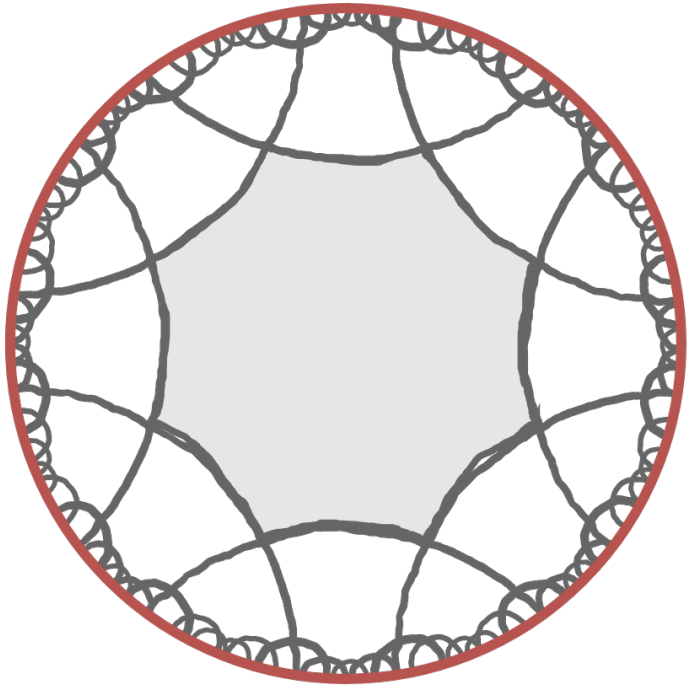
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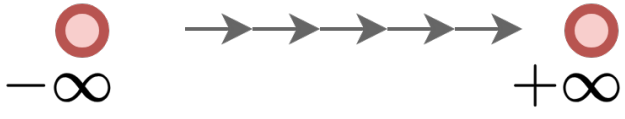
Hyperbolic groups



$\partial F \cong$ Cantor set



$\partial \pi_1 \Sigma \cong S^1$

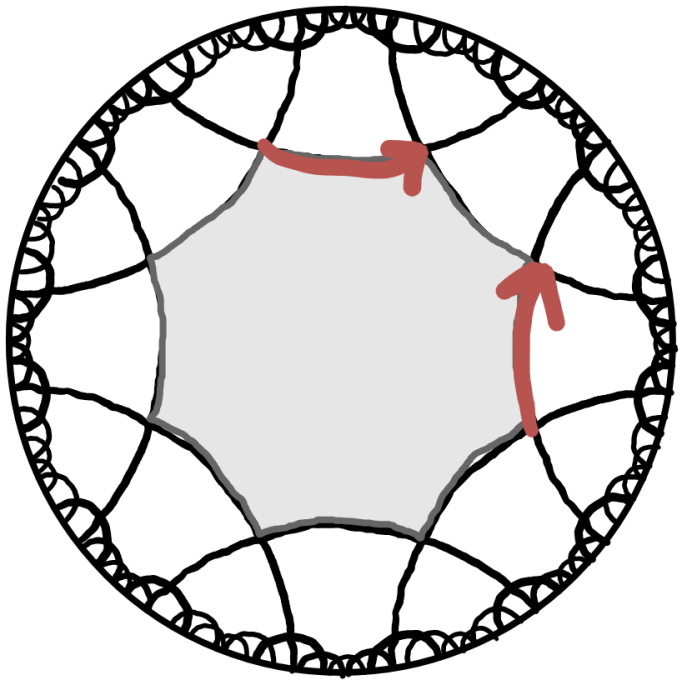


$\partial \mathbb{Z} \cong \{+\infty, -\infty\}$

Fuchsian and quasi-Fuchsian representations

Fuchsian representations are discrete and faithful homomorphisms

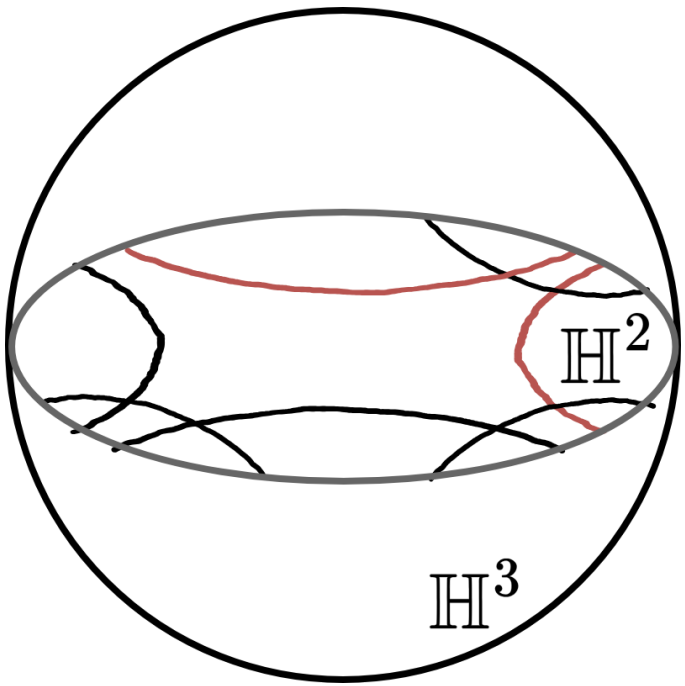
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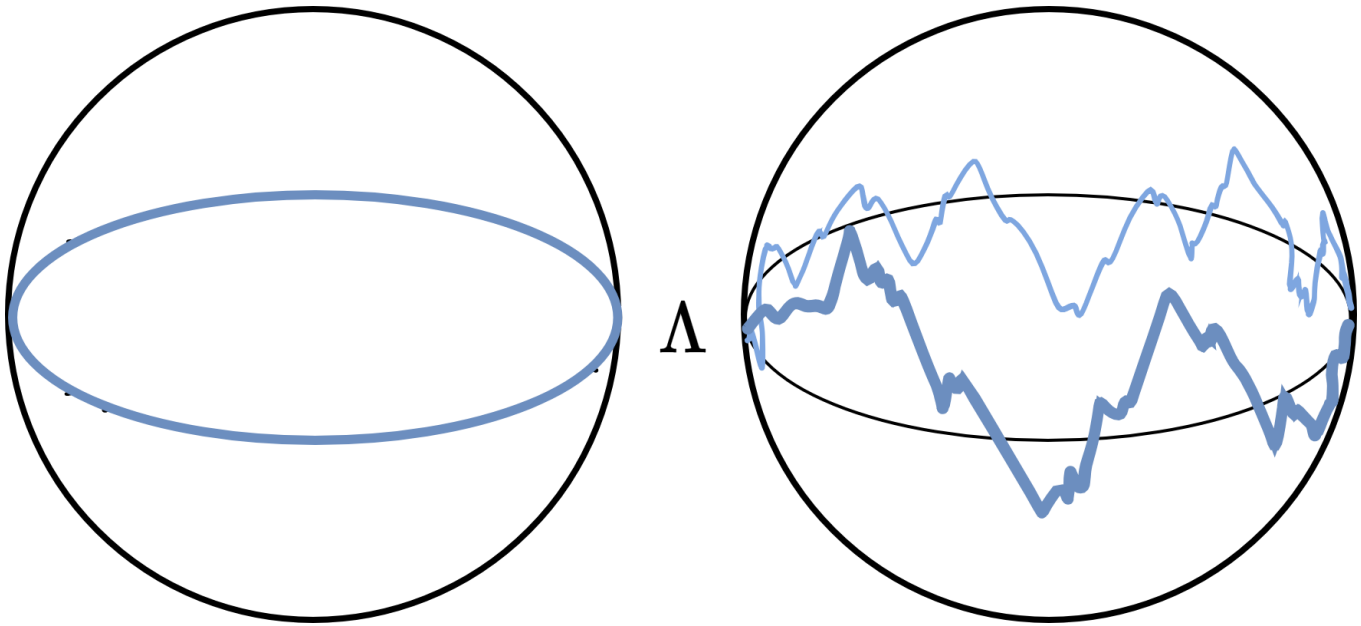
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Quasi-Fuchsian representations are perturbations of ρ_F in $\mathrm{Isom}^+(\mathbb{H}^3)$

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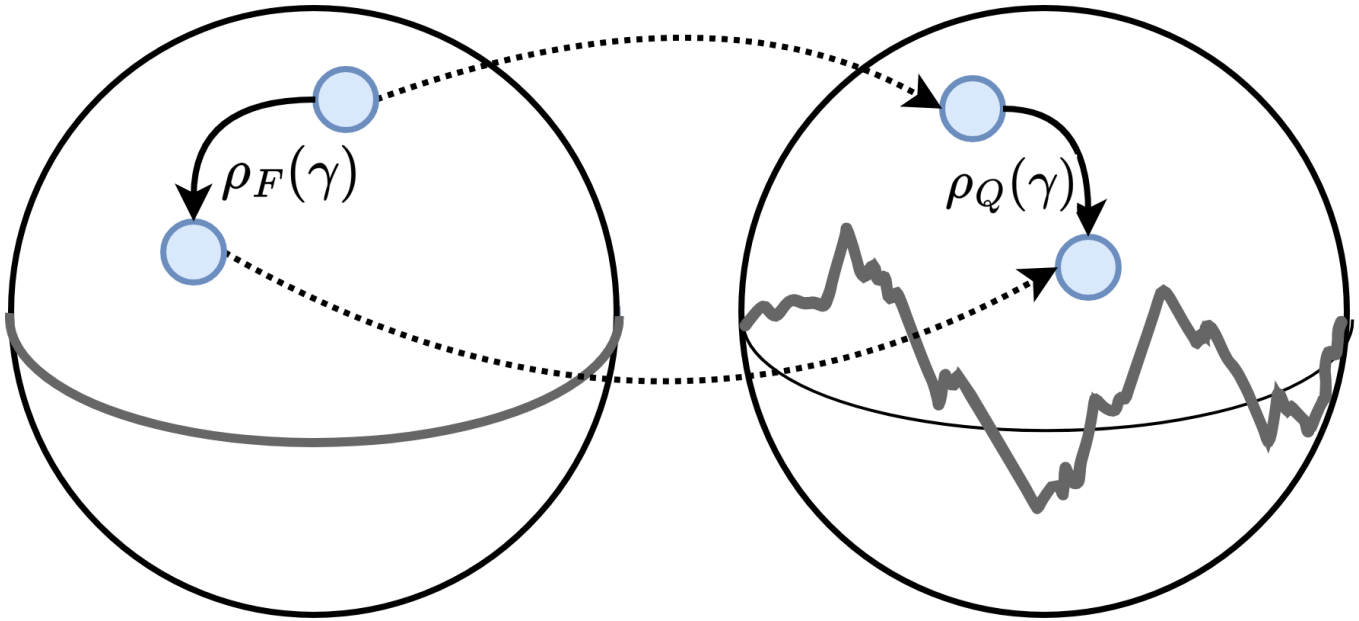
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Theorem (Sullivan '82): Fuchsian and quasi-Fuchsian representations are topologically conjugate.



Anosov representations

Definition (Labourie '06, GGKW '17):

Suppose a representation $\rho : \Gamma \to G$ admits a continuous, ρ-equivariant boundary map $\xi : \partial\Gamma \to \mathcal{F}$ such that

- ξ is *transverse*,
- ξ is *dynamics-preserving*, and
- ρ is Δ*-expanding*.

Then ρ is *Anosov*.

The set of Anosov representations in $\mathrm{Hom}(\Gamma, G)$ is open.

Hyperconvex representations

Definition (PSW '20, V Chapter 4):

Let $\rho : \Gamma \to G$ Anosov with limit set $\mathrm{im}\xi = \Lambda \subset \mathcal{F}$.

Suppose for all distinct $x, y, z \in \Lambda$ that

$$\mathrm{Rich}^0_{x,y} \cap \mathrm{Rich}^0_{x,z} = \{x\}.$$

Then ρ is *hyperconvex*.

The set of hyperconvex representations in $\mathrm{Hom}(\Gamma, G)$ is open.

Anosov components in $SL_3\mathbb{R}$

We can build higher-rank Anosov representations from Fuchsian representations $\rho_F : \pi_1\Sigma \to PSL_2\mathbb{R}$.

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Hitchin representations

Real perturbations of

$$\rho_H = \iota_3 \circ \rho_F$$

Anosov (Labourie '06)

Hyperconvex (Hitchin '91)

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Anosov (Labourie '06)

Hyperconvex (Hitchin '91)

Barbot representations

Real Anosov perturbations of

$$\rho_B = \begin{pmatrix} \rho_F & \\ & 1 \end{pmatrix}$$

Anosov (Barbot '05)

Not hyperconvex (Guichard '08)

Anosov components in $SL_3\mathbb{C}$

The $\mathbb{C}$-*Hitchin* and $\mathbb{C}$-*Barbot* representations are complex Anosov perturbations of ρ_H and ρ_B.

Question: Are they distinct $\mathbb{C}$-Anosov components?

- Same Hom component. (Li '93)
- So far, same topological invariants. (Hart '26)
- Possibly distinguished by hyperconvexity?
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Goal: find a relationship between hyperconvexity and the dynamics of Anosov representations.

Theorem B: Topological conjugacies preserve hyperconvexity

Topological conjugacies

Let $\rho, \rho' : \Gamma \to G$ be Anosov representations.
Then $\rho(\Gamma)$ and $\rho'(\Gamma)$ act on $\mathcal{F} \cong G/B$ by left multiplication.

Definition: Let $F : \mathcal{F} \to \mathcal{F}$ be a homeomorphism such that for all $\gamma \in \Gamma$ and $x \in \mathcal{F}$ we have

$$F(\rho(\gamma) \cdot x) = \rho'(\gamma) \cdot F(x).$$

Then F is a *topological conjugacy* between ρ and ρ'.

Theorem B: Topological conjugacies preserves hyperconvexity

Theorem B (V Chapter 5):

Let ρ, ρ' be topologically conjugate.

Then ρ is hyperconvex if and only if ρ' is hyperconvex.

Lemma B: F ϕ-matches the Richardson varieties of Λ

Because ξ and ξ' are dynamics-preserving, then for all $x \in \Lambda$,

$$F(x) = \xi' \circ \xi^{-1}(x) \in \Lambda'.$$

Thus we use the notation $F(x) = x' \in \Lambda'$.

Lemma B: Let F be a topological conjugacy between ρ and ρ'. Then there exists a Bruhat order automorphism $\phi \in \mathrm{Aut}(W)$ such that F ϕ-matches the Richardson varieties of Λ.

For all distinct $x, y \in \Lambda$,

$$F(\alpha_u(x) \cap \alpha_v(y)) = \alpha_{\phi(u)}(x') \cap \alpha_{\phi(v)}(y').$$

Proof of Theorem B

Theorem B: $\rho \sim \rho'$ implies ρ hyperconvex iff ρ' hyperconvex

Let F be a topological conjugacy between ρ and ρ'.
Suppose ρ is hyperconvex.

Let $x', y', z' \in \Lambda'$ distinct. Then $\exists x, y, z \in \Lambda$ such that $F(x) = x'$, etc.

Since ρ is hyperconvex, then

$$\mathrm{Rich}^0_{x,y} \cap \mathrm{Rich}^0_{x,z} = \{x\}.$$

By Lemma B, $F(\mathrm{Rich}^0_{x,y}) = \mathrm{Rich}^0_{x',y'}$ and $F(\mathrm{Rich}^0_{x,z}) = \mathrm{Rich}^0_{x',z'}$.

Since F is a homeomorphism, then

$$\begin{aligned}\mathrm{Rich}^0_{x',y'} \cap \mathrm{Rich}^0_{x',z'} &= F(\mathrm{Rich}^0_{x,y}) \cap F(\mathrm{Rich}^0_{x,z}) \\ &= F(\mathrm{Rich}^0_{x,y} \cap \mathrm{Rich}^0_{x,z}) \\ &= F(\{x\}) = \{x'\}.\end{aligned}$$

Thus ρ' is hyperconvex.

Proof of Lemma B

Lemma B: *F ϕ-matches the Richardson varieties of Λ.*

Let $\gamma \in \Gamma$ be infinite order. Then $\rho(\gamma)$ is proximal in G, so it has attracting fixed point $x = \xi(\gamma^+)$ and repelling fixed point $y = \xi(\gamma^-)$.

By Corollary A, the fixed points of $\rho(\gamma)$ and $\rho'(\gamma)$ are the Richardson 0-varieties of x, y and x', y'. Thus $F(\mathrm{Rich}^0_{x,y}) = \mathrm{Rich}^0_{x',y'}$.

To prove $F(\mathrm{Rich}^0_{x,y}) = F(\mathrm{Rich}^0_{x',y'})$ for any distinct $x, y \in \Lambda$, we use the

- Continuity of Richardson varieties,
- Density of $(\gamma^+, \gamma^-) \in \partial\Gamma \times \partial\Gamma$, and
- Convergence group action of Γ on $\partial\Gamma$.

Since F preserves dynamics and the Richardson 0-varieties, then by Theorem A it preserves the Bruhat order on $\mathrm{Rich}_{x,y}$ via $\phi_{xy} \in \mathrm{Aut}(W)$.

There is enough continuity that ϕ_{xy} doesn't depend on x or y.

Warning: existence of topological conjugacies

In general, it is difficult to construct nontrivial topological conjugacies. (E.g. $\rho \sim \rho$ or $\rho \sim g\rho g^{-1}$.)

Question:

Is there an equivalence relation on Anosov representations that

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Answer: Yes.

Theorem C: l_Δ-topological conjugacies preserve hyperconvexity

Bruhat ideal thickening

Let $I \subset W$ be a Bruhat ideal, and let $x \in \mathcal{F}$.
We can "thicken" a flag by taking a union of Schubert varieties:

$$Th^I(x) = \bigcup_{w \in I} \alpha_w(x)$$

We can "thicken" the limit set Λ of an Anosov representation:

$$Th^I(\Lambda) = \bigcup_{x \in \Lambda} Th^I(x)$$

Then $\rho(\Gamma)$ acts on $Th^I(\Lambda)$.

Partial topological conjugacy

Let ρ, ρ' be Anosov representations, and let $I \subset W$ be a Bruhat ideal.

Definition: Suppose there exists a homeomorphism $F : Th^I(\Lambda) \to Th^I(\Lambda')$ such that

$$F(\rho(\gamma) \cdot x = \rho'(\gamma) \cdot F(x)$$

for all $x \in Th^I(\Lambda)$ and $\gamma \in \Gamma$. Then F is an *I-topological conjugacy* between ρ and ρ'.

E.g. all Anosov representations are $\{e\}$-topologically conjugate.

E.g. a W-topological conjugacy is a topological conjugacy.

Theorem C: I_Δ-topological conjugacies preserve hyperconvexity

Let $I_\Delta = \{e, s_1, \ldots, s_n\} \subset W$.

Theorem C (V Chapter 5):

Let ρ, ρ' be I_Δ-topologically conjugate.

Then ρ is hyperconvex if and only if ρ' is hyperconvex.

Theorem C: I_Δ-topological conjugacies preserve hyperconvexity

Let $I_\Delta = \{e, s_1, \ldots, s_n\} \subset W$.

Theorem C (V Chapter 5):

Let ρ, ρ' be I_Δ-topologically conjugate.

Then ρ is hyperconvex if and only if ρ' is hyperconvex.

We construct nontrivial I_Δ-topological conjugacies.

Proof of Theorem C

Similar argument to Theorem B.

Let $x_k^y := x_{s_k}^y$ and $\alpha_k(x) := \alpha_{s_k}(x)$ for any $1 \leq k \leq n$.

Proposition: If for all $1 \leq k \leq n$ we have $x_k^y \notin \mathrm{Rich}_{x,y}^0 \cap \mathrm{Rich}_{x,z}^0$, then

$$\mathrm{Rich}_{x,y}^0 \cap \mathrm{Rich}_{x,z}^0 = \{x\}.$$

Lemma: Suppose F is an I_Δ-topological conjugacy between ρ and ρ'. Then there is a permutation $\sigma \in \mathrm{Sym}(n)$ such that F σ-matches the Richardson varieties x_k^y and $\alpha_k(x)$ of $x \in \Lambda$.

$$F(x_k^y) = x'^{y'}_{\sigma(k)}$$
$$F(\alpha_k(x)) = \alpha_{\sigma(k)}(x').$$

Existence of I_Δ-topological conjugacies

Theorem (FPV '24): If Γ admits a hyperconvex representation, then Γ is virtually isomorphic to a Kleinian group.

E.g. $\Gamma \cong \pi_1\Sigma$.
E.g. $\partial\Gamma \cong S^2$.

Proposition (V Chapter 5): Let $\rho, \rho' : \pi_1\Sigma \to SL_n\mathbb{C}$ be quasi-Fuchsian.

Then ρ and ρ' are I_Δ-topologically conjugate.

Proposition (V Chapter 5): Let Γ such that $\partial\Gamma \cong S^2$.

Let $\rho, \rho' : \Gamma \to G$ be hyperconvex representations.

Then ρ and ρ' are I_Δ-topologically conjugate.

Future directions

Conjecture: Hyperconvex representations are I_Δ-top conjugate, but not fully top conjugate except in rank 1.

Idea: use quasi-symmetric maps from FPV '24 to construct.

If so, and if top conjugacy classes of irreducible Anosov representations are open, then the $\mathbb{C}$-Hitchin and $\mathbb{C}$-Barbot Anosov components are disjoint.

Future directions

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If so, and if top conjugacy classes of irreducible Anosov representations are open, then the $\mathbb{C}$-Hitchin and $\mathbb{C}$-Barbot Anosov components are disjoint.

Another goal: understand the incidence geometry description of the Richardson varieties and flag varieties of exceptional Lie groups.

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